

MEASURABLE EUCLIDEAN RAMSEY THEORY FOR LINES AND FINITE CONFIGURATIONS

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ABSTRACT. Euclidean Ramsey theory asks which finite point configurations must appear monochromatically in every coloring of Euclidean space, a question that remains open in many simple cases such as equally spaced points on a line. Building on the Fourier-analytic and linear-programming approach to measurable distance-avoiding sets initiated by Székely and developed by de Oliveira Filho and Vallentin, we extend these methods to a range of Euclidean Ramsey problems when the color classes are assumed to be measurable. For finite configurations $X_1, \dots, X_r \subset \mathbb{E}^d$, write $\mathbb{E}^d \rightarrow_m (X_1, \dots, X_r)$ if every measurable r -coloring of $\mathbb{E}^d$ contains an isometric copy of X_i in color i for some i , and let ℓ_m be the unit-spaced m -point line. Our main result is $\mathbb{E}^2 \rightarrow_m (\ell_2, \ell_7)$, whereas for arbitrary colorings the best known result is $\mathbb{E}^2 \rightarrow (\ell_2, \ell_5)$. We also prove $\mathbb{E}^2 \rightarrow_m (\ell_2, \ell_2, \ell_3)$ and the higher-dimensional results $\mathbb{E}^3 \rightarrow_m (\ell_3, \ell_4)$, $\mathbb{E}^4 \rightarrow_m (\ell_3, \ell_5)$, $\mathbb{E}^7 \rightarrow_m (\ell_3, \ell_6)$, and $\mathbb{E}^4 \rightarrow_m (\ell_4, \ell_4)$. Beyond lines, we show that in any measurable two-coloring of $\mathbb{E}^2$ in which one color contains no pair at unit distance, the other contains both a regular unit pentagon and a unit square together with its center. Finally, we prove that every measurable two-coloring of $\mathbb{E}^3$ contains either a monochromatic unit square or a monochromatic noncoplanar tetrahedron with four unit distances and two distances $\sqrt{2}$, giving a partial result under measurability on the question of whether every two-coloring of $\mathbb{E}^3$ contains a monochromatic unit square.

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1. INTRODUCTION

Euclidean Ramsey theory, initiated in a series of papers by Erdős, Graham, Montgomery, Rothschild, Spencer, and Straus, asks which finite configurations are unavoidable in every coloring of Euclidean space [16, 17, 18]. For two configurations $X, Y \subset \mathbb{E}^d$, we write $\mathbb{E}^d \rightarrow (X, Y)$ if every red–blue coloring of $\mathbb{E}^d$ contains either a red X or a blue Y , up to isometry. Its negation, denoted $\mathbb{E}^d \nrightarrow (X, Y)$, means that there exists a red–blue coloring with no red X and no blue Y . More generally, for finite configurations $X_1, \dots, X_r \subset \mathbb{E}^d$, write $\mathbb{E}^d \rightarrow (X_1, \dots, X_r)$ if every r -coloring of $\mathbb{E}^d$ contains an isometric copy of X_i in color i for some i . Thus $\mathbb{E}^d \nrightarrow (X_1, \dots, X_r)$ means that there exists an r -coloring in which color i contains no copy of X_i for every i . We write $\rightarrow_m$ when the color classes are required to be Lebesgue measurable. This restriction asks the question for a smaller class of colorings, but it also makes spatial averaging and Fourier analysis available. The indicator function of each color class can be averaged over large balls, and its Fourier transform controls averages of products at fixed distances. The arbitrary-coloring and measurable questions therefore have different strengths and different available methods.

We first discuss line configurations. For $m \geq 2$, let $\ell_m = \{0, 1, \dots, m-1\} \subset \mathbb{R}$ be the unit-spaced m -point line. Erdős, Graham, Montgomery, Rothschild, Spencer, and Straus proved $\mathbb{E}^2 \rightarrow (\ell_2, \ell_4)$ and, more generally, $\mathbb{E}^2 \rightarrow (\ell_2, K)$ for every three-point configuration $K \subset \mathbb{E}^2$ [17]. Tsaturian later proved $\mathbb{E}^2 \rightarrow (\ell_2, \ell_5)$ [29]. In dimension 3, Iván proved $\mathbb{E}^3 \rightarrow (\ell_2, K)$ for every five-point configuration $K \subset \mathbb{E}^3$, and Arman and Tsaturian proved $\mathbb{E}^3 \rightarrow (\ell_2, \ell_6)$ [21, 3]. On the other hand, Szlam proved that for some constant $c' > 0$, $\mathbb{E}^d \rightarrow (\ell_2, K)$ for every $K \subset \mathbb{E}^d$ with $|K| \leq 2^{c'd}$ [28]. In the opposite direction, Conlon and Fox proved $\mathbb{E}^2 \nrightarrow (\ell_2, \ell_{10^{10}})$ and, for some constant $c > 0$, $\mathbb{E}^d \nrightarrow (\ell_2, \ell_m)$ whenever $d \geq 2$ and $m \geq 2^{cd}$ [6]. Currier, Mody, Xie, and Zhang improved the planar obstruction to $\mathbb{E}^2 \nrightarrow (\ell_2, \ell_{6330})$ [10]. The known planar results therefore leave the gap $6 \leq m \leq 6329$. We improve the lower bound in the measurable setting.

Theorem 1. $\mathbb{E}^2 \rightarrow_m (\ell_2, \ell_7)$.

For three colors, the Moser spindle shows that the unit-distance graph of the plane requires at least four colors, and hence $\mathbb{E}^2 \rightarrow (\ell_2, \ell_2, \ell_2)$ [24]. Currier, Moore, and Yip proved the three obstructions $\mathbb{E}^d \nrightarrow (\ell_3, \ell_3, \ell_8)$, $\mathbb{E}^d \nrightarrow (\ell_3, \ell_4, \ell_7)$, and $\mathbb{E}^d \nrightarrow (\ell_3, \ell_5, \ell_5)$ for every $d \geq 2$ [12]. We prove a related result for three colors.

Theorem 2. $\mathbb{E}^2 \rightarrow_m (\ell_2, \ell_2, \ell_3)$.

For two colors and red ℓ_3 , Erdős, Graham, Montgomery, Rothschild, Spencer, and Straus proved $\mathbb{E}^3 \rightarrow (K, K)$ for every three-point configuration $K \subset \mathbb{E}^3$, and hence $\mathbb{E}^3 \rightarrow (\ell_3, \ell_3)$ [16]. Currier, Moore, and Yip later proved the positive planar result $\mathbb{E}^2 \rightarrow (\ell_3, \ell_3)$ with computer assistance [11]. Conlon and Wu proved $\mathbb{E}^d \nrightarrow (\ell_3, \ell_{10^{50}})$ for every $d \geq 2$ [8]. Führer and Töth improved 10^{50} to 1177 [19], and Currier, Moore, and Yip later improved it to 20 [12]. Thus, in the plane, there remains a gap $4 \leq m \leq 19$. We prove three results for measurable colorings in higher dimensions. The dimensions in

these statements record where the present constructions work. Their optimal dimensions remain open.

Theorem 3. $\mathbb{E}^3 \rightarrow_m (\ell_3, \ell_4)$.

Theorem 4. $\mathbb{E}^4 \rightarrow_m (\ell_3, \ell_5)$.

Theorem 5. $\mathbb{E}^7 \rightarrow_m (\ell_3, \ell_6)$.

For red ℓ_4 , Erdős, Graham, Montgomery, Rothschild, Spencer, and Straus proved $\mathbb{E}^d \nrightarrow (\ell_6, \ell_6)$ and $\mathbb{E}^d \nrightarrow (\ell_4, \ell_4, \ell_4)$ for every $d \geq 2$ [16]. For two colors, interchanging the colors in $\mathbb{E}^2 \rightarrow (\ell_2, \ell_4)$ gives $\mathbb{E}^2 \rightarrow (\ell_4, \ell_2)$ [17]. Currier, Moore, and Yip proved $\mathbb{E}^d \nrightarrow (\ell_4, \ell_{14})$ and $\mathbb{E}^d \nrightarrow (\ell_5, \ell_8)$ for every $d \geq 2$ [12]. Thus the planar gap for red ℓ_4 is $3 \leq m \leq 13$. In a higher dimension we prove a result for measurable colorings.

Theorem 6. $\mathbb{E}^4 \rightarrow_m (\ell_4, \ell_4)$.

More generally, Conlon and Führer proved that if a finite configuration $X \subset \mathbb{E}^d$ is not contained in a sphere, then there is an integer m such that $\mathbb{E}^n \nrightarrow (X, \ell_m)$ for every $n \geq d$ [7].

We next turn to finite point configurations. For $n \geq 3$, let P_n be the regular n -gon of side length one. For $s > 0$, let $W_n(s)$ be a regular n -gon of side length s together with its center. In particular, $W_4(1)$ is a unit square together with its center.

Juhász proved that $\mathbb{E}^2 \rightarrow (\ell_2, K)$ for every four-point configuration $K \subset \mathbb{E}^2$, and in the same paper gave a twelve-point configuration $K \subset \mathbb{E}^2$ with $\mathbb{E}^2 \nrightarrow (\ell_2, K)$ [22]. Csizmadia and Tóth later reduced the size of such an obstruction to eight. $\mathbb{E}^2 \nrightarrow (\ell_2, W_7(\frac{9}{5} \sin(\pi/7)))$ [9]. Thus $\mathbb{E}^2 \rightarrow (\ell_2, W_3(s))$ for every $s > 0$, while the corresponding statement fails for $W_7(\frac{9}{5} \sin(\pi/7))$. More broadly, for $n \geq 3$ and $s > 0$, one can ask when $\mathbb{E}^2 \rightarrow (\ell_2, W_n(s))$, where $W_n(s)$ is a regular n -gon of side length s together with its center. This problem has two parameters, namely the number of sides n and the side length s . We begin with the measurable case.

Theorem 7. $\mathbb{E}^2 \rightarrow_m (\ell_2, W_4(1))$.

Note that this is a measurable result, whereas the Csizmadia–Tóth result holds for arbitrary colorings.

For regular polygons, Juhász’s four-point result gives $\mathbb{E}^2 \rightarrow (\ell_2, P_4)$. The case P_6 is elementary and needs no measurability. The next case immediately beyond P_4 is P_5 , which we prove under the measurable hypothesis.

Theorem 8. $\mathbb{E}^2 \rightarrow_m (\ell_2, P_5)$.

If e_1, e_2, e_3 are orthonormal, set $\mathcal{S} = \{0, e_1, e_2, e_1 + e_2\}$ and $\mathcal{T} = \{0, e_1, e_2, \frac{1}{2}e_1 + \frac{\sqrt{3}}{2}e_3\}$. Thus $\mathcal{S}$ is a unit square and $\mathcal{T}$ is a noncoplanar tetrahedron with four distances 1 and two distances $\sqrt{2}$.

For the final configuration, a folklore construction gives a two-coloring of $\mathbb{E}^2$ with no monochromatic copy of $\mathcal{S}$, that is, $\mathbb{E}^2 \nrightarrow (\mathcal{S}, \mathcal{S})$. See Gasarch, Gezalyan, and Parker [20]. Erdős, Graham, Montgomery, Rothschild, Spencer, and Straus proved $\mathbb{E}^6 \rightarrow (\mathcal{S}, \mathcal{S})$ [16]. As Cantwell notes, a trivial modification of their proof gives $\mathbb{E}^5 \rightarrow (\mathcal{S}, \mathcal{S})$ [5]. Cantwell

later proved $\mathbb{E}^4 \rightarrow (\mathcal{S}, \mathcal{S})$ for arbitrary colorings [5]. Thus $\mathbb{E}^d \rightarrow (\mathcal{S}, \mathcal{S})$ is known for every $d \geq 4$. Dimension 3 remains open, while dimension 2 is ruled out by the coloring above. We address this missing dimension with the following measurable result. Up to isometry, four points with four pairwise distances equal to 1 and two equal to $\sqrt{2}$ form either a square, when the two $\sqrt{2}$ -pairs are disjoint, or the noncoplanar tetrahedron $\mathcal{T}$, when the two pairs share a vertex. We prove that one of these two configurations must occur monochromatically.

Theorem 9. *Every measurable two-coloring of $\mathbb{E}^3$ contains a monochromatic copy of either $\mathcal{S}$ or $\mathcal{T}$.*

The nine results play different roles. Theorem 1 extends the planar positive results from ℓ_5 to a measurable result at ℓ_7 , while Theorem 2 gives a three-color analogue. The higher-dimensional line theorems show that the same method applies beyond the planar setting. Theorem 7 closes the planar centered-square target $W_4(1)$, while Theorem 8 gives the corresponding measurable result for the regular pentagon. Theorem 9 gives a measurable partial result toward the unresolved three-dimensional square problem.

Our method extends the Fourier-analytic and linear-programming approach to measurable distance-avoiding sets, initiated by Székely and developed by de Oliveira Filho and Vallentin [27, 14]. Related work treats finite unit-distance constraints, correlations involving three or more points, and more general optimization problems [4, 23, 1, 2, 13].

The spherical Fourier representation and the positivity of the associated measures are standard ingredients in measurable distance-avoidance arguments. We combine them with finite inequalities for Ramsey statements in which different color classes are subject to different avoidance conditions. In the planar arguments, these inequalities involve several color indicators at once. In higher dimensions, we organize the possible local color patterns into a finite directed graph. Assigning a number to each vertex makes the intermediate terms cancel when the local inequalities are added along a line. This gives one method for treating all of the results below.

Our three-color result gives a simple example of the argument. Call the three colors red, blue, and green, and consider the 3×3 triangular-lattice finite configuration used to prove Theorem 2. Suppose that this finite configuration contains neither a red pair nor a blue pair at distance one, and no green copy of ℓ_3 . We assign coefficients to the points and to pairs of points at the distances appearing in the finite configuration. The resulting quadratic functional $\mathcal{L}$ is chosen so that every coloring of the finite configuration with these avoidance properties satisfies $\mathcal{L} \geq 0$. The linear terms record the colors of individual points, and the quadratic terms record pairs of points with prescribed colors.

Now place this finite configuration at all translations and rotations and average $\mathcal{L}$ over a large ball. The boundary contribution tends to zero as the ball grows. The remaining terms are color densities and pair correlations at the finitely many distances in the configuration. After the densities are subtracted from the indicator functions, Fourier analysis writes each correlation as an integral against a nonnegative measure. The integrand is a finite linear combination of spherical averages determined by the distances in the configuration. A pointwise upper bound for this integrand therefore gives an upper bound for the averaged value of $\mathcal{L}$. In the nine-point case this upper bound

is negative, contradicting the nonnegative lower bound obtained from every coloring of the finite configuration. The coefficients are chosen so that the local expression is nonnegative while the associated polynomial in the color density and kernel bound give a negative average. This links the finite combinatorial inequality to the Fourier estimate for its average.

The other planar results use the same argument with different finite configurations, quadratic expressions, and coefficients. For the line results, a possible coloring of a short sequence of consecutive points is treated as a local pattern. Appending one new point shifts the sequence by one place and gives a directed edge between two allowed patterns. Assign a number $\phi(u)$ to each pattern u . If the edge from u to v has contribution $W(u, v)$, the local inequalities take the form $W(u, v) + \phi(u) - \phi(v) \geq 0$. Summing these inequalities along an oriented line makes the intermediate ϕ -terms cancel and leaves only contributions from the two ends. This is the line analogue of the finite-configuration inequality. For $W_4(1)$, P_5 , and the square–tetrahedron result, the local object is instead a finite point set containing copies of the configurations under consideration.

The planar and line averaging propositions later in the paper separate the finite lower bound, the averaged upper bound, and the boundary error in these two forms of the argument.

The coefficients are found by linear programs whose constraints encode the finite colorings of the configuration and the desired bounds for the Fourier kernels at selected frequencies. The linear programs are used to find the coefficients. The proofs use rounded rational or integer values, verify the local inequalities directly, and establish the kernel bounds analytically over the full frequency range.

The paper is organized as follows. Section 2 develops the spherical averaging estimates. Sections 3–8 prove the nine Ramsey theorems. Section 9 describes the linear programs, and Section 10 gives finite-radius consequences and open problems. The appendices record the finite data and the rational verifications used in the proofs.

2. FOURIER AVERAGING

2.1. Spherical averaging. Let $d \geq 2$ and let $B_T = \{x \in \mathbb{R}^d : |x| \leq T\}$ be the closed Euclidean ball of radius T centered at the origin. We use $|B_T|$ for its volume. Let $h : \mathbb{R}^d \rightarrow \mathbb{R}$ be bounded and measurable, and let $h_T = h\mathbf{1}_{B_T}$ be its restriction to B_T . Thus h_T is zero outside the ball.

Let σ_{d-1} be surface measure on the unit sphere $\mathbb{S}^{d-1}$ divided by the total surface area, so that $\sigma_{d-1}(\mathbb{S}^{d-1}) = 1$. For $\omega \in \mathbb{S}^{d-1}$, write ω_1 for its first coordinate.

For $r \geq 0$, define the spherical autocorrelation of h_T at distance r by

$$\mathcal{A}_{h,T}(r) = \frac{1}{|B_T|} \int_{\mathbb{S}^{d-1}} \int_{\mathbb{R}^d} h_T(y) h_T(y + r\omega) \, dy \, d\sigma_{d-1}(\omega),$$

The product compares the values of h_T at two points whose displacement is $r\omega$. The outer integral averages over all directions of that displacement. The factor $|B_T|^{-1}$ normalizes the average by the volume of the ball. The truncation to B_T makes all these integrals finite.

We next express this correlation in terms of the Fourier transform. We use the unitary convention

$$\widehat{f}(\xi) = \frac{1}{(2\pi)^{d/2}} \int_{\mathbb{R}^d} e^{-i\xi \cdot x} f(x) dx,$$

for which Plancherel's identity has no additional constant [26, Eq. (6.5.1), Theorem 6.5.6, and Eq. (6.5.33)]. The vector ξ is the Fourier frequency. It records the oscillation of the function $x \mapsto e^{i\xi \cdot x}$. For $a \in \mathbb{R}^d$, Plancherel's identity and the Fourier transform of a translate gives

$$\int_{\mathbb{R}^d} h_T(y) h_T(y+a) dy = \int_{\mathbb{R}^d} e^{-i\xi \cdot a} |\widehat{h_T}(\xi)|^2 d\xi.$$

The complex exponential is the factor by which the Fourier component at frequency ξ changes under translation by a .

Plancherel gives $\int_{\mathbb{R}^d} |\widehat{h_T}(\xi)|^2 d\xi = \int_{B_T} |h(x)|^2 dx < \infty$. Define the resulting finite nonnegative measure $\widehat{\nu}_{h,T}$ on $\mathbb{R}^d$ by

$$d\widehat{\nu}_{h,T}(\xi) = \frac{|\widehat{h_T}(\xi)|^2}{|B_T|} d\xi.$$

It is nonnegative because its density is nonnegative. Let $\nu_{h,T}$ be the measure on $[0, \infty)$ obtained by recording only the length of the frequency: for every Borel set $B \subset [0, \infty)$, $\nu_{h,T}(B) = \widehat{\nu}_{h,T}(\{\xi : |\xi| \in B\})$. Thus the variable t below is the magnitude $|\xi|$ of a Fourier frequency. The measure changes with T because h_T changes with T , but t always ranges over $[0, \infty)$ and has this same meaning. From this point on, $\nu_{h,T}$ denotes a measure on radial Fourier variables of this kind, while μ is reserved for the centered red-indicator measure defined below.

When $a = r\omega$ and we average over all directions ω , rotational invariance shows that the average of $e^{-i\xi \cdot a}$ depends only on the product $r|\xi|$. We denote this one-variable function

$$\Omega_d(u) = \int_{\mathbb{S}^{d-1}} e^{-iu\omega_1} d\sigma_{d-1}(\omega).$$

Therefore

$$\int_{\mathbb{S}^{d-1}} e^{-ir\xi \cdot \omega} d\sigma_{d-1}(\omega) = \Omega_d(r|\xi|),$$

because, when $\xi \neq 0$, a rotation carries $\xi/|\xi|$ to the first coordinate vector. When $\xi = 0$, both sides are 1. The function Ω_d is real and even by symmetry, and $\Omega_d(0) = 1$ because σ_{d-1} is a probability measure.

For $\alpha \geq 0$ and $t \geq 0$, let Γ denote the gamma function and let J_α be the Bessel function of the first kind with parameter α , defined by

$$J_\alpha(t) = \sum_{n=0}^{\infty} \frac{(-1)^n}{n! \Gamma(n + \alpha + 1)} \left(\frac{t}{2}\right)^{2n + \alpha}.$$

The series converges for every t , and in particular $J_0(t) = \sum_{n=0}^{\infty} (-1)^n (t^2/4)^n / (n!)^2$.

Lemma 10. *Let $d \geq 2$. For $t > 0$,*

$$\Omega_d(t) = \Gamma\left(\frac{d}{2}\right) \left(\frac{2}{t}\right)^{d/2-1} J_{d/2-1}(t) = \sum_{n=0}^{\infty} \frac{(-1)^n t^{2n}}{2^n n! d(d+2) \cdots (d+2n-2)},$$

where $J_{d/2-1}$ is the Bessel function of the first kind with parameter $d/2 - 1$, and the product in the denominator is 1 when $n = 0$. Moreover, for every $t \in \mathbb{R}$,

$$|\Omega'_d(t)| \leq \frac{2\Gamma(d/2)}{(d-1)\sqrt{\pi}\Gamma((d-1)/2)}.$$

Proof. The Bessel representation is the spherical Fourier-transform formula [14, Eq. (1)]. Substituting the power series for $J_{d/2-1}$ [25, Eq. 10.2.2] and simplifying the gamma factors gives, with $\alpha = d/2 - 1$,

$$\Gamma\left(\frac{d}{2}\right)\left(\frac{2}{t}\right)^\alpha \frac{(-1)^n t^{2n+\alpha}}{2^{2n+\alpha} n! \Gamma(n+d/2)} = \frac{(-1)^n t^{2n}}{2^n n! \prod_{j=0}^{n-1} (d+2j)},$$

which is the series above.

For the derivative estimate, dominated convergence applies because $|\partial_t e^{-it\omega_1}| = |\omega_1| \leq 1$. Therefore

$$\Omega'_d(t) = -i \int_{\mathbb{S}^{d-1}} \omega_1 e^{-it\omega_1} d\sigma_{d-1}(\omega),$$

so $|\Omega'_d(t)|$ is at most the mean of $|\omega_1|$ on the sphere. The first coordinate of a uniformly distributed point of $\mathbb{S}^{d-1}$ has density

$$\frac{\Gamma(d/2)}{\sqrt{\pi}\Gamma((d-1)/2)}(1-u^2)^{(d-3)/2}$$

on $[-1, 1]$ [15, Remark 2.10]. Consequently,

$$\begin{aligned} \int_{\mathbb{S}^{d-1}} |\omega_1| d\sigma_{d-1}(\omega) &= \frac{2\Gamma(d/2)}{\sqrt{\pi}\Gamma((d-1)/2)} \int_0^1 u(1-u^2)^{(d-3)/2} du \\ &= \frac{2\Gamma(d/2)}{(d-1)\sqrt{\pi}\Gamma((d-1)/2)}. \end{aligned}$$

With $v = 1 - u^2$ and $dv = -2u du$, one has

$$\int_0^1 u(1-u^2)^{(d-3)/2} du = \frac{1}{2} \int_0^1 v^{(d-3)/2} dv = \frac{1}{d-1},$$

which gives the final equality. $\square$

The cases needed below are

$$\begin{aligned} \Omega_2(t) &= J_0(t), & \Omega_3(t) &= \frac{\sin t}{t}, \\ \Omega_4(t) &= \frac{2J_1(t)}{t}, & \Omega_7(t) &= 15 \frac{(3-t^2)\sin t - 3t\cos t}{t^5}. \end{aligned}$$

Each expression has value 1 at $t = 0$ by continuity. These formulas follow from the preceding lemma and the standard half-integer formulas for $J_{1/2}$ and $J_{5/2}$ [25, Eq. (10.47.3) and Sec. 10.49(i)].

Lemma 11. *Let $\nu_{h,T}$ be the measure on radial Fourier variables obtained from $\hat{\nu}_{h,T}$ by mapping ξ to $|\xi|$. Then, for every $r \geq 0$,*

$$\mathcal{A}_{h,T}(r) = \int_0^\infty \Omega_d(rt) d\nu_{h,T}(t).$$

Moreover,

$$\nu_{h,T}([0, \infty)) = \frac{1}{|B_T|} \int_{B_T} h(x)^2 dx.$$

Thus $\nu_{h,T}$ has mass $|B_T|^{-1} \|h_T\|_2^2$.

Proof. This is the spherical autocorrelation identity from [14, Sec. 2]. We give the proof to fix the normalization. Since $h_T \in L^2(\mathbb{R}^d)$ is real-valued, Plancherel's identity and the Fourier transform of a translate give, for every $a \in \mathbb{R}^d$,

$$\int_{\mathbb{R}^d} h_T(y) h_T(y+a) dy = \int_{\mathbb{R}^d} e^{-i\xi \cdot a} |\widehat{h_T}(\xi)|^2 d\xi.$$

Set $a = r\omega$ and average over $\omega \in \mathbb{S}^{d-1}$. Fubini's theorem applies because $|\widehat{h_T}|^2$ is integrable. Rotational invariance and the definition of Ω_d give

$$\int_{\mathbb{S}^{d-1}} e^{-ir\xi \cdot \omega} d\sigma_{d-1}(\omega) = \Omega_d(r|\xi|).$$

Grouping the integral according to the value $t = |\xi|$, which is the definition of $\nu_{h,T}$, gives

$$\frac{1}{|B_T|} \int_{\mathbb{R}^d} \Omega_d(r|\xi|) |\widehat{h_T}(\xi)|^2 d\xi = \int_0^\infty \Omega_d(rt) d\nu_{h,T}(t).$$

Finally, Plancherel's identity gives

$$\nu_{h,T}([0, \infty)) = \frac{1}{|B_T|} \int_{\mathbb{R}^d} |\widehat{h_T}(\xi)|^2 d\xi = \frac{1}{|B_T|} \int_{B_T} h(x)^2 dx.$$

□

The measure $\nu_{h,T}$ is nonnegative, so an upper or lower bound for the function multiplying it in the integral gives the corresponding bound after the total mass of the measure is known. We call that multiplying function the kernel. In dimension two, write $\omega = (\cos \theta, \sin \theta)$ and let ρ_θ denote rotation through θ . Since $\Omega_2 = J_0$, the kernel associated with two points at distance r is $J_0(rt)$.

2.2. Truncation estimates. Lemma 11 integrates over all of $\mathbb{R}^d$, whereas our averages are taken over B_T . The difference is confined to a boundary region whose relative volume tends to zero. If $a \in \mathbb{R}^d$ is fixed and $T > |a|$, then $B_T \triangle (B_T + a)$, the set of points belonging to exactly one of the two balls, is contained in $B_{T+|a|} \setminus B_{T-|a|}$. Hence

$$|B_T \triangle (B_T + a)| = O(T^{d-1}). \quad (2.1)$$

The containing annulus has volume $O(T^{d-1})$ for fixed a , since $|B_r| = |B_1|r^d$. The constants implicit in $O(T^{d-1})$ may depend on the fixed distances, the dimension, and the bound for the function, while remaining independent of T .

For quantitative statements, put $D_d = d2^d$. If $T \geq k \geq 0$ and $|a| = k$, then

$$\frac{|B_T \triangle (B_T + a)|}{|B_T|} \leq \frac{(T+k)^d - (T-k)^d}{T^d} \leq \frac{D_d k}{T}. \quad (2.2)$$

The last inequality follows from the mean value theorem and $T+k \leq 2T$.

For a bounded measurable $f : \mathbb{R}^d \rightarrow \mathbb{R}$, define its average on B_T by $p_T(f) = |B_T|^{-1} \int_{B_T} f(x) dx$. When f is an indicator function, $p_T(f)$ is the proportion of the

corresponding set in B_T . For $k \geq 0$, define the finite-ball spherical two-point correlation at distance k by

$$C_{k,T}(f) = \frac{1}{|B_T|} \int_{\mathbb{S}^{d-1}} \int_{B_T} f(x) f(x + k\omega) dx d\sigma_{d-1}(\omega).$$

Thus p_T records a one-point average, whereas $C_{k,T}$ records a two-point correlation. The quantity $\mathcal{A}_{h,T}$ above is the whole-space autocorrelation of the truncated function h_T . The estimates below compare $C_{k,T}(f)$ with this Fourier-analytic quantity. For a fixed function, we abbreviate these quantities to p_T and $C_{k,T}$.

For the next lemma, fix a bounded measurable function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ and write $B = \|f\|_\infty$. For its second part, also fix $m \geq 1$ and a measurable function $\phi : \mathbb{R}^m \rightarrow \mathbb{R}$ bounded on $[-\|f\|_\infty, \|f\|_\infty]^m$. Let M_ϕ be the supremum of $|\phi|$ on this cube.

Lemma 12. *The following estimates hold.*

(i) For fixed $k \geq 0$,

$$C_{k,T}(f) = \int_0^\infty \Omega_d(kt) d\nu_{f,T}(t) + O(T^{-1}).$$

If $T \geq k$, the error has absolute value at most $B^2 D_d k / T$.

(ii) The average over $x \in B_T$ and $\omega \in \mathbb{S}^{d-1}$ of

$$\phi(f(x), \dots, f(x + (m-1)\omega)) - \phi(f(x + \omega), \dots, f(x + m\omega))$$

is $O(T^{-1})$. If $T \geq 1$, the absolute value of this average is at most $M_\phi D_d / T$.

Proof. (i) For a fixed ω , the autocorrelation of the truncation $f \mathbf{1}_{B_T}$ is

$$\int_{\mathbb{R}^d} (f \mathbf{1}_{B_T})(x) (f \mathbf{1}_{B_T})(x + k\omega) dx = \int_{B_T \cap (B_T - k\omega)} f(x) f(x + k\omega) dx.$$

The two domains being compared are $B_T \cap (B_T - k\omega)$ and B_T , respectively. The absolute difference is at most $B^2 |B_T \Delta (B_T - k\omega)|$. Averaging over ω , dividing by $|B_T|$, applying (2.2), and then using Lemma 11 prove both the first formula and its explicit error bound.

(ii) For the second statement, put $\Phi(x, \omega) = \phi(f(x), \dots, f(x + (m-1)\omega))$. The stated average is

$$\frac{1}{|B_T|} \int_{\mathbb{S}^{d-1}} \int_{B_T} (\Phi(x, \omega) - \Phi(x + \omega, \omega)) dx d\sigma_{d-1}(\omega).$$

For fixed ω , the substitution $y = x + \omega$ gives

$$\int_{B_T} \Phi(x + \omega, \omega) dx = \int_{B_T + \omega} \Phi(y, \omega) dy.$$

Thus the absolute difference between the two integrals is at most $M_\phi |B_T \Delta (B_T + \omega)|$, uniformly in ω . After the translation, both integrals have the same integrand $\Phi(\cdot, \omega)$, so their difference is supported only on the symmetric difference of the two domains. Averaging, dividing by $|B_T|$, and applying (2.2) with $k = 1$ proves the claim and its explicit bound. $\square$

For the next lemma, work in dimension two. Fix a measurable function $h : \mathbb{R}^2 \rightarrow [-1, 1]$ and points $p, q \in \mathbb{R}^2$, and put $M = \max\{|p|, |q|\}$. Assume $T > 2M$. Let $\nu_{h,T}$ be the measure on radial Fourier variables defined above.

Lemma 13. *The following two estimates hold:*

(i)

$$\left| \frac{1}{2\pi|B_T|} \int_0^{2\pi} \int_{B_T} h(x + \rho_\theta p) dx d\theta - p_T(h) \right| \leq \frac{4M}{T}, \quad (2.3)$$

(ii)

$$\left| \frac{1}{2\pi|B_T|} \int_0^{2\pi} \int_{B_T} h(x + \rho_\theta p) h(x + \rho_\theta q) dx d\theta - \int_0^\infty J_0(|p - q|t) d\nu_{h,T}(t) \right| \leq \frac{12M}{T}. \quad (2.4)$$

Proof. (i) If $a \in \mathbb{R}^2$ and $|a| < T$, then the annulus estimate is

$$|B_T \triangle (B_T + a)| \leq \pi(T + |a|)^2 - \pi(T - |a|)^2 = 4\pi T|a|.$$

For (2.3), substitute $y = x + \rho_\theta p$. The domains $B_T + \rho_\theta p$ and B_T have symmetric difference of area at most $4\pi T|p| \leq 4\pi TM$. Since $|h| \leq 1$, this bounds the difference of the corresponding integrals. Averaging over θ and dividing by $|B_T| = \pi T^2$ gives (2.3).

(ii) For (2.4), put $w_\theta = \rho_\theta(q - p)$ and again substitute $y = x + \rho_\theta p$. The inner integral becomes

$$\int_{B_T + \rho_\theta p} h(y) h(y + w_\theta) dy,$$

whereas the truncated autocorrelation is

$$\int_{\mathbb{R}^2} h_T(y) h_T(y + w_\theta) dy = \int_{B_T \cap (B_T - w_\theta)} h(y) h(y + w_\theta) dy.$$

The integrand has absolute value at most 1. By the triangle inequality for symmetric differences, the difference of the two integrals is therefore at most

$$|B_T \triangle (B_T + \rho_\theta p)| + |B_T \triangle (B_T - w_\theta)| \leq 4\pi T|p| + 4\pi T|w_\theta|.$$

Since $|w_\theta| = |q - p| \leq 2M$, the last expression is at most $12\pi TM$. This bound is the source of the factor $12M/T$. After averaging and dividing by πT^2 , the error is at most $12M/T$. As θ varies uniformly, w_θ varies uniformly over the circle of radius $|p - q|$. Lemma 11 and $\Omega_2 = J_0$ identify its truncated autocorrelation with the integral in (2.4). □

For indicator functions, subtracting the average separates the constant term from the remaining correlation. This gives the form used in the higher-dimensional proofs.

For the next lemma, let $r : \mathbb{R}^d \rightarrow \{0, 1\}$ be measurable, put $p = p_T(r)$, and define the T -dependent centered function

$$f_T = r - p_T(r).$$

Let $\mu_{f,T}$ be the measure on radial Fourier variables associated with the centered function $f_T \mathbf{1}_{B_T}$. Then $\mu_{f,T}$ is nonnegative with mass $p_T(r)(1 - p_T(r))$. For a measurable signed

coloring $s : \mathbb{R}^d \rightarrow \{-1, 1\}$, let $\nu_{s,T}$ be the measure on radial Fourier variables associated with s_T .

Lemma 14. *For every fixed $k \geq 0$,*

$$C_{k,T}(r) = p^2 + \int_0^\infty \Omega_d(kt) d\mu_{f,T}(t) + O(T^{-1}).$$

Moreover, $\nu_{s,T}$ is nonnegative with mass 1 and satisfies

$$C_{k,T}(s) = \int_0^\infty \Omega_d(kt) d\nu_{s,T}(t) + O(T^{-1}).$$

For $T \geq k$, the error in the first formula has absolute value at most $2D_d k/T$, and the error in the second formula has absolute value at most $D_d k/T$.

Proof. The measure $\mu_{f,T}$ is nonnegative by definition. Lemma 11 and the identity $r^2 = r$ show that its mass is

$$\frac{1}{|B_T|} \int_{B_T} (r - p)^2 dx = p - 2p^2 + p^2 = p(1 - p).$$

Lemma 12 applied to f_T gives $C_{k,T}(f_T) = \int_0^\infty \Omega_d(kt) d\mu_{f,T}(t) + E_{k,T}$, where $|E_{k,T}| \leq D_d k/T$ when $T \geq k$. To express the left side in terms of r , define

$$A_{k,T} = \frac{1}{|B_T|} \int_{\mathbb{S}^{d-1}} \int_{B_T} r(x + k\omega) dx d\sigma_{d-1}(\omega).$$

When $k = 0$, $A_{0,T} = p$ and both identities below are exact, with zero boundary error. Assume $k > 0$ for the remaining discussion. Expanding $(r(x) - p)(r(x + k\omega) - p)$ gives a term $C_{k,T}(r)$. The average of the first factor is $p_T(r) = p$, so the constant terms cancel and

$$C_{k,T}(f_T) = C_{k,T}(r) - pA_{k,T}.$$

For each ω , translation by $k\omega$ changes the domain of integration from B_T to $B_T + k\omega$. Equation (2.2) and $0 \leq r \leq 1$ therefore give $|A_{k,T} - p| \leq D_d k/T$. Hence $C_{k,T}(f_T) = C_{k,T}(r) - p^2 + E'_{k,T}$, where $|E'_{k,T}| \leq D_d k/T$. Combining this with the error $E_{k,T}$ gives the stated $2D_d k/T$ bound and proves the first formula.

For s , use the radial measure $\nu_{s,T}$ associated with s_T and apply Lemmas 11 and 12 to s . The mass of this measure is 1 because $s^2 = 1$, and the explicit error bound is $D_d k/T$. $\square$

The planar expressions are finite sums of one-point terms and products of two values of a coloring. The next proposition averages every such expression at once. Its kernel is obtained by replacing each product at distance r by $J_0(rt)$.

For the next proposition, let $P \subset \mathbb{R}^2$ be finite, with $|p| \leq M$ for every $p \in P$, and let $h_1, \dots, h_s : \mathbb{R}^2 \rightarrow [-1, 1]$ be measurable. In every sum indexed by $\{p, q\} \subset P$, the pairs are unordered two-element subsets of P , so each pair occurs once. For real coefficients c , $\alpha_{j,p}$, and $\beta_{j,pq} = \beta_{j,qp}$, define

$$\mathcal{L}_{x,\theta} := c + \sum_{j=1}^s \sum_{p \in P} \alpha_{j,p} h_j(x + \rho_\theta p) + \sum_{j=1}^s \sum_{\{p,q\} \subset P} \beta_{j,pq} h_j(x + \rho_\theta p) h_j(x + \rho_\theta q).$$

Put $K_j(t) = \sum_{\{p,q\} \subset P} \beta_{j,pq} J_0(|p-q|t)$. For $T > 2M$, let A_T be the average of $\mathcal{L}_{x,\theta}$ over $B_T \times [0, 2\pi]$. It is given by

$$A_T := \frac{1}{2\pi|B_T|} \int_0^{2\pi} \int_{B_T} \mathcal{L}_{x,\theta} \, dx \, d\theta.$$

The measure $\nu_{h_j,T}$ on radial Fourier variables is obtained from $\widehat{\nu}_{h_j,T}$ by mapping ξ to $|\xi|$.

Proposition 15. *With this notation, if $T > 2M$, then*

$$\left| A_T - c - \sum_{j=1}^s \left(\sum_{p \in P} \alpha_{j,p} \right) p_T(h_j) - \sum_{j=1}^s \int_0^\infty K_j(t) \, d\nu_{h_j,T}(t) \right| \leq \frac{4M}{T} \sum_{j,p} |\alpha_{j,p}| + \frac{12M}{T} \sum_{j,\{p,q\}} |\beta_{j,pq}|.$$

Proof. For each j and $p \in P$, equation (2.3) gives

$$\frac{1}{2\pi|B_T|} \int_0^{2\pi} \int_{B_T} h_j(x + \rho_\theta p) \, dx \, d\theta = p_T(h_j) + \varepsilon_{j,p},$$

where $|\varepsilon_{j,p}| \leq 4M/T$. Thus the total contribution of the linear terms is

$$\sum_{j=1}^s \left(\sum_{p \in P} \alpha_{j,p} \right) p_T(h_j) + E_1.$$

The error satisfies $|E_1| \leq (4M/T) \sum_{j,p} |\alpha_{j,p}|$. For each j and each unordered pair $\{p, q\} \subset P$, equation (2.4) similarly gives

$$\frac{1}{2\pi|B_T|} \int_0^{2\pi} \int_{B_T} h_j(x + \rho_\theta p) h_j(x + \rho_\theta q) \, dx \, d\theta = \int_0^\infty J_0(|p-q|t) \, d\nu_{h_j,T}(t) + \varepsilon_{j,pq},$$

where $|\varepsilon_{j,pq}| \leq 12M/T$. It follows that the total contribution of the quadratic terms is

$$\sum_{j=1}^s \sum_{\{p,q\} \subset P} \beta_{j,pq} \int_0^\infty J_0(|p-q|t) \, d\nu_{h_j,T}(t) + E_2,$$

where $|E_2| \leq (12M/T) \sum_{j,\{p,q\}} |\beta_{j,pq}|$. The sums are finite, so they may be interchanged with the integrals. For each j , the resulting integrand is $\sum_{\{p,q\} \subset P} \beta_{j,pq} J_0(|p-q|t) = K_j(t)$. Adding the constant, linear, and quadratic contributions gives the stated formula. Its error is $E_1 + E_2$, the sum of the linear and quadratic truncation errors, and the triangle inequality gives the claimed bound. $\square$

For the next proposition, retain the notation of Proposition 15. Suppose an argument on a finite configuration shows that every allowed coloring of P satisfies $\mathcal{L} \geq c_0$, and that every translated and rotated copy of P in a measurable global coloring avoiding the target configurations is allowed. Put

$$A_j = \sum_{p \in P} \alpha_{j,p}, \quad m_{j,T} = \nu_{h_j,T}([0, \infty)),$$

The boundary constant in Proposition 15 is

$$C_{\text{bd}} = 4M \sum_{j,p} |\alpha_{j,p}| + 12M \sum_{j,\{p,q\}} |\beta_{j,pq}|.$$

Assume that there are numbers $\kappa_1, \dots, \kappa_s$ and $\eta > 0$ such that $K_j(t) \leq \kappa_j$ for every j and $t \geq 0$, and that the density and Fourier masses arising from the global coloring satisfy

$$c + \sum_{j=1}^s A_j p_T(h_j) + \sum_{j=1}^s \kappa_j m_{j,T} \leq c_0 - \eta$$

for every $T > 2M$.

Proposition 16. *Under these assumptions, no such global coloring exists. More specifically, the contradiction is obtained for any*

$$T > \max \left\{ 2M, \frac{C_{\text{bd}}}{\eta} \right\}.$$

Proof. The local hypothesis gives $A_T \geq c_0$. Since the measures $\nu_{h_j,T}$ are nonnegative, the kernel bounds give

$$\sum_{j=1}^s \int_0^\infty K_j(t) d\nu_{h_j,T}(t) \leq \sum_{j=1}^s \kappa_j m_{j,T}.$$

Proposition 15 therefore gives $A_T \leq c_0 - \eta + C_{\text{bd}}/T < c_0$, a contradiction. $\square$

It is useful to normalize each finite inequality at zero. If the finite argument gives $\mathcal{L} \geq c_0$, replace it by

$$\hat{\mathcal{L}} = \mathcal{L} - c_0, \quad \hat{A}_T = A_T - c_0.$$

Then every allowed local coloring has $\hat{\mathcal{L}} \geq 0$, and an averaged upper bound of the form $A_T \leq c_0 - \eta + O(T^{-1})$ becomes

$$\hat{A}_T \leq -\eta + O(T^{-1}).$$

The subsequent finite inequalities use this normalization whenever the local lower bound is positive.

2.3. Analytic estimates. The preceding results reduce the later proofs to pointwise estimates for finite linear combinations of the functions $\Omega_d(rt)$. We record here the elementary bounds used outside bounded intervals. On bounded intervals, the later proofs combine the derivative estimate in Lemma 10 with values sampled at finitely many points.

Lemma 17. *The following estimates hold:*

- (i) $|J'_0(x)| < 2/3$ for every $x \in \mathbb{R}$.
- (ii) If $x \geq 2$, then $|J_0(x)| \leq \sqrt{2/(\pi x)}$.
- (iii) If $q \geq 1$ and $t \geq 25$, then $|J_0(\sqrt{q}t)| < 1/(6q^{1/4})$.
- (iv) If $q \geq 1$ and $t \geq 100$, then $|J_0(\sqrt{q}t)| < 5/(6\sqrt{t}q^{1/4})$.
- (v) If $t > 0$, then $|\Omega_3(t)| \leq 1/t$.
- (vi) If $t > 0$, then $|\Omega_4(t)| \leq 2/t$.

(vii) If $t \geq 10$, then $|\Omega_7(t)| \leq 30/t^3$.

Proof. The integral representation $J_0(x) = \pi^{-1} \int_0^\pi \cos(x \sin \theta) d\theta$ [25, Eq. 10.9.1] may be differentiated under the integral sign, so $|J'_0(x)| \leq 2/\pi < 2/3$.

For (ii), put $\chi = x - \pi/4$. The expansion and signed remainder theorem in [25, Eq. 10.17.3 and Sec. 10.17(iii)] give

$$J_0(x) = \sqrt{\frac{2}{\pi x}} \left(A(x) \cos \chi + B(x) \sin \chi \right),$$

where

$$A = 1 - \frac{9}{128x^2} + R_e, \quad B = \frac{1}{8x} - R_o,$$

and $0 \leq R_e \leq 11025/(98304x^4)$, $0 \leq R_o \leq 225/(3072x^3)$. For $x \geq 2$, one has $0 \leq A \leq 1 - 9/(128x^2) + 11025/(98304x^4)$ and $0 \leq B \leq 1/(8x)$. Thus it suffices to verify that the squares of these two upper bounds sum to at most 1 for $x \geq 2$. Cauchy–Schwarz now gives (ii).

Substituting $x = \sqrt{q}t$ into (ii) gives $|J_0(\sqrt{q}t)| \leq q^{-1/4} \sqrt{2/(\pi t)}$. Using $\pi > 3$ gives (iii) and (iv). The formula $\Omega_3(t) = \sin t/t$ gives (v), while the integral representation of J_1 [25, Eq. 10.9.2] gives $|J_1(t)| \leq 1$ and hence (vi). Finally, $|\Omega_7(t)| \leq 15(t^2 + 3t + 3)/t^5 \leq 30/t^3$ for $t \geq 10$. $\square$

For the verifications on bounded intervals below, write $u = z^2/4$. If the series for $J_0(z)$ is truncated after the term with index N and $u < (N+2)^2$, then the successive tail terms have ratio at most $u/(N+2)^2$. Thus the omitted tail satisfies

$$\left| \sum_{m=N+1}^{\infty} \frac{(-1)^m u^m}{(m!)^2} \right| \leq \frac{u^{N+1}}{((N+1)!)^2} \frac{1}{1 - u/(N+2)^2}. \quad (2.5)$$

We will repeatedly use the following consequence of the mean value theorem: if $|K'| \leq D$ on an interval and every point of the interval lies within $h/2$ of a sample point where $K \leq M$, then $K \leq M + Dh/2$ throughout the interval.

At a rational sample point t , the series parameter $u = qt^2/4$ is rational whenever the squared distance q is rational, so the partial sum and the right side of (2.5) are rational. When q is irrational, we enclose it by rational endpoints and propagate the interval through the series with outward rounding. The coefficient sign then selects the endpoint used in an upper or lower kernel bound. The derivative estimates in the corresponding proof extend the sample-point bounds between sample points. For the two planar configurations $W_4(1)$ and P_5 , the corresponding computer verifications are summarized in Appendix A. That appendix gives the finite sets of colorings and line-graph vertices, arithmetic with rational numbers, interval bounds with rational endpoints, and bounds for the omitted series terms used in those verifications.

3. PROOF OF THEOREM 1

3.1. The finite planar inequality. Let $e_1, e_2 \in \mathbb{R}^2$ be unit vectors satisfying $e_1 \cdot e_2 = 1/2$. For $i \in \{0, 1\}$ and $0 \leq j \leq 8$, put $v_{ij} = (i - 1/2)e_1 + (j - 4)e_2$, and set

$P = \{v_{ij} : i \in \{0, 1\}, 0 \leq j \leq 8\}$. We use the label ij for v_{ij} . The formula gives $|v_{ij}| \leq \sqrt{73}/2$ for every vertex, which will be used in Section 3.2.

For $i \in \{0, 1\}$ and $s \in \{0, 1, 2\}$, put $P_{i,s} = \{v_{ij} : s \leq j \leq s + 6\}$. These are the six copies of ℓ_7 used below, and they are all the copies of ℓ_7 contained in P : a unit-spaced progression in the triangular lattice has label difference $(\Delta i, \Delta j)$ satisfying $\Delta i^2 + \Delta i \Delta j + \Delta j^2 = 1$, whose six possibilities are $(\pm 1, 0), (0, \pm 1), (1, -1), (-1, 1)$. Since $i \in \{0, 1\}$, a progression of seven points must have $\Delta i = 0$ and $|\Delta j| = 1$, so it is one of the sets $P_{i,s}$.

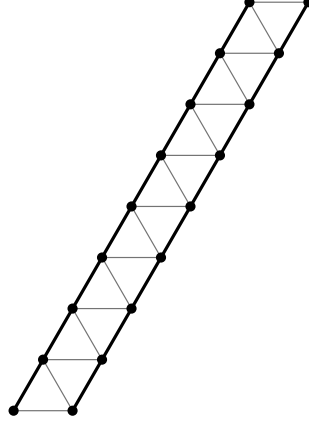


FIGURE 1. The eighteen-point triangular-lattice strip P . All drawn edges have unit length.

For a coloring $\text{Col} : P \rightarrow \{R, B\}$, let r_p be the red indicator and put $b_p = 1 - r_p$. Define

$$N_R = \sum_{p \in P} r_p, \quad N_B = \sum_{p \in P} b_p.$$

For each vertex $p \in P$ and unordered pair $\{p, w\} \subset P$, let c_p and $a_{pw} = a_{wp}$ be the integer coefficients in the local inequality. The coefficients were chosen with a linear program and rounded to integers. Lemma 18 verifies the resulting inequality. Define

$$V_R = \sum_{p \in P} c_p r_p, \quad E_R = \sum_{\{p, w\} \subset P} a_{pw} r_p r_w.$$

The eighteen values c_p and the 153 values a_{pw} are listed in Appendix B, so $\mathcal{L}$ is explicitly defined.

Define

$$\mathcal{L} = -1001 + \frac{500}{9} N_R + \frac{500}{9} N_B + V_R + E_R, \quad (3.1)$$

If the configuration has no red unit edge, then every pair term corresponding to a unit pair vanishes. Also, $N_R + N_B = 18$, so the first three terms in (3.1) are $-1001 + (500/9) \cdot 18 = -1$. Since the remaining coefficients and indicators are integers, $\mathcal{L}$ is integer-valued.

Lemma 18. *If a coloring of P has no red unit edge and none of the six sets $P_{i,s}$ is entirely blue, then $\mathcal{L} \geq 0$.*

Proof. For a subset $R \subseteq P$, let Col_R be the coloring whose red set is R , and write $r_p(R) = \mathbf{1}_R(p)$. Define

$$F(R) = -1 + \sum_{p \in P} c_p r_p(R) + \sum_{\{p,q\} \subset P} a_{pq} r_p(R) r_q(R).$$

For Col_R , one has $N_R = |R|$ and $N_B = 18 - |R|$, so the first three terms in (3.1) combine to -1 , and $F(R)$ is the value of $\mathcal{L}$ on Col_R .

The unit pairs in the labeling $00, 01, \dots, 08, 10, 11, \dots, 18$ are

$$\begin{aligned} &\{v_{ij}, v_{i,j+1}\} \text{ for } i \in \{0, 1\}, 0 \leq j \leq 7, \\ &\{v_{0j}, v_{1j}\} \text{ for } 0 \leq j \leq 8, \\ &\{v_{0j}, v_{1,j-1}\} \text{ for } 1 \leq j \leq 8. \end{aligned}$$

Let $\mathcal{A}_7$ be the family of subsets $R \subseteq P$ such that no unit pair is contained in R and $R \cap P_{i,s} \neq \emptyset$ for $i \in \{0, 1\}$ and $s \in \{0, 1, 2\}$. The first condition excludes red unit pairs, and the second excludes blue copies of ℓ_7 . Thus it suffices to verify $F(R) \geq 0$ for $R \in \mathcal{A}_7$.

Enumerating $\mathcal{A}_7$ leaves 751 subsets. Substitution of the integer coefficients from Appendix B gives $F(R) \geq 0$ for every $R \in \mathcal{A}_7$, which proves the lemma. $\square$

3.2. Averaging the local expression. To compute the average of $\mathcal{L}$, we now group its pair terms according to squared distance. For integer pairs $p = (i, j)$ and $q = (i', j')$, write $Q(p - q) = (i - i')^2 + (i - i')(j - j') + (j - j')^2$.

Lemma 19. *For vertices labelled by $p, q \in \{0, 1\} \times \{0, 1, \dots, 8\}$, one has $|v_p - v_q|^2 = Q(p - q)$.*

Proof. Write $\Delta i = i - i'$ and $\Delta j = j - j'$. Since $|e_1| = |e_2| = 1$ and $e_1 \cdot e_2 = 1/2$, one has $|v_p - v_q|^2 = (\Delta i)^2 + \Delta i \Delta j + (\Delta j)^2 = Q(p - q)$. $\square$

Suppose that a measurable coloring $\text{COL} : \mathbb{R}^2 \rightarrow \{R, B\}$ contains no red ℓ_2 and no blue ℓ_7 . Let r be the red indicator and put $b = 1 - r$. For $x \in \mathbb{R}^2$ and $\theta \in \mathbb{R}$, the coloring induced on P is $\text{Col}_{x,\theta} = (\text{COL}(x + \rho_\theta v))_{v \in P}$. For $T > \sqrt{73}$, set

$$A_T = \frac{1}{2\pi|B_T|} \int_0^{2\pi} \int_{B_T} \mathcal{L}(\text{Col}_{x,\theta}) dx d\theta.$$

Every induced coloring satisfies Lemma 18, so $A_T \geq 0$.

Define

$$\begin{aligned} H_R(t) = & 681 - 489J_0(t) + 1035J_0(\sqrt{3}t) - 669J_0(2t) \\ & + 600J_0(\sqrt{7}t) - 700J_0(3t) + 548J_0(\sqrt{13}t) \\ & - 355J_0(4t) + 441J_0(\sqrt{21}t) - 105J_0(5t) \\ & + 339J_0(\sqrt{31}t) - 114J_0(6t) - 139J_0(\sqrt{43}t) \\ & + 16J_0(7t) - 90J_0(\sqrt{57}t), \end{aligned} \tag{3.2}$$

$$H_B(t) = 1000. \tag{3.3}$$

The distance-class data underlying this grouping are recorded in Appendix A.2.

Proposition 20. *For every $T > \sqrt{73}$,*

$$\left| A_T + 1001 - \int_0^\infty H_R(t) d\nu_{r,T}(t) - \int_0^\infty H_B(t) d\nu_{b,T}(t) \right| \leq \frac{54734\sqrt{73}}{T}. \quad (3.4)$$

Moreover,

$$\nu_{r,T}([0, \infty)) + \nu_{b,T}([0, \infty)) = 1. \quad (3.5)$$

Proof. The finite configuration has radius $M_0 = \sqrt{73}/2$. From the coefficients in Appendix B, the absolute sums of the linear and quadratic coefficients are 1681 and 8562. Proposition 15 therefore gives the error

$$\frac{4M_0}{T} 1681 + \frac{12M_0}{T} 8562 = \frac{54734\sqrt{73}}{T}.$$

The linear coefficient sums are 681 for red and 1000 for blue. The blue part has no quadratic terms. Grouping the coefficients for red pairs by squared distance gives (3.2), so Proposition 15 gives (3.4). Finally, the two measures have masses $p_T(r)$ and $p_T(b)$, and $r + b = 1$, which gives (3.5). $\square$

3.3. Bounds for the kernel. Put $K = H_R - 1000$. The bounded-interval calculation used below is recorded in Appendix A.2.

Proposition 21. *For every $t \geq 0$, one has $H_R(t) < 1000$.*

Proof. Put $K = H_R - 1000$. Appendix A.2 gives rational upper bounds for K on $[0, 100]$, together with a derivative bound that extends them between the sample points. Hence $K < 0$ on this interval.

For $t \geq 100$, the same appendix gives $\sum_q |A_q| q^{-1/4} < 3600$. Lemma 17 therefore yields

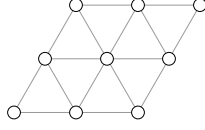
$$H_R(t) \leq 681 + \frac{5}{6\sqrt{t}} \sum_q \frac{|A_q|}{q^{1/4}} < 681 + 300 = 981 < 1000.$$

$\square$

Proof of Theorem 1. Choose $T > 54734\sqrt{73}$. This choice also satisfies $T > \sqrt{73}$, so Lemma 18 and Proposition 20 give the following bound. This is the planar averaging argument of Proposition 16. The local lemma supplies the lower bound, while the Fourier proposition supplies the averaged upper bound.

$$0 \leq A_T \leq -1001 + \int_0^\infty H_R(t) d\nu_{r,T}(t) + \int_0^\infty H_B(t) d\nu_{b,T}(t) + \frac{54734\sqrt{73}}{T}.$$

Proposition 21, (3.3), and (3.5), together with positivity of the two measures, imply $A_T \leq -1 + \frac{54734\sqrt{73}}{T} < 0$, contradicting $A_T \geq 0$. $\square$

FIGURE 2. The nine-point triangular-lattice configuration Λ .

4. PROOF OF THEOREM 2

4.1. A quadratic expression on a nine-point configuration. Let $e_1, e_2 \in \mathbb{R}^2$ be unit vectors with $e_1 \cdot e_2 = 1/2$. Put $v_{ij} = (i-1)e_1 + (j-1)e_2$ for $0 \leq i, j \leq 2$, and let $\Lambda = \{v_{ij} : 0 \leq i, j \leq 2\}$. We label v_{ij} by ij .

For integer pairs $p = (i, j)$ and $q = (i', j')$, put $Q(p - q) = (i - i')^2 + (i - i')(j - j') + (j - j')^2$. Then $|v_{ij} - v_{i'j'}|^2 = Q((i, j) - (i', j'))$. The nonzero squared distances in Λ are 1, 3, 4, 7, 12. The copies of ℓ_3 used below lie in the three unit directions $(1, 0)$, $(0, 1)$, and $(1, -1)$.

Let $\mathcal{G}$ be generated by $\sigma(i, j) = (j, i)$ and $\tau(i, j) = (2 - i, 2 - j)$. It has four elements and preserves the configuration and its distances. Its vertex orbits are

representative	orbit	size
00	$\{00, 22\}$	2
01	$\{01, 10, 12, 21\}$	4
02	$\{02, 20\}$	2
11	$\{11\}$	1.

The action on unordered pairs has thirteen orbits. Let $\mathcal{O}_V$ and $\mathcal{O}_P$ denote the vertex and pair orbits, respectively. For $U \in \mathcal{O}_P$, let $d(U)$ be the common squared distance of the pairs in U .

For a coloring with colors G, R, B , define $g = 1_G$ and $s = 1_R - 1_B$. Thus $g^2 = g$, $s^2 = 1 - g$, and $gs = 0$. For a vertex orbit O and a pair orbit U , put $[g]_O = |O|^{-1} \sum_{v \in O} g(v)$ and

$$[gg]_U = \frac{1}{|U|} \sum_{\{v,w\} \in U} g(v)g(w),$$

$$[ss]_U = \frac{1}{|U|} \sum_{\{v,w\} \in U} s(v)s(w).$$

A red-blue swap fixes g and changes the sign of s . At one vertex, the relations $g^2 = g$, $s^2 = 1 - g$, and $gs = 0$ reduce every quadratic monomial to a linear combination of 1, g , and s . Averaging over the swap removes s . For two distinct vertices, the same swap removes the terms containing one factor s , leaving only gg and ss . Averaging over $\mathcal{G}$ then makes each remaining coefficient constant on a vertex or pair orbit.

Define

$$\mathcal{L}(\text{Col}) = -196 + \sum_{O \in \mathcal{O}_V} c_O [g]_O + \sum_{U \in \mathcal{O}_P} (a_U [gg]_U + b_U [ss]_U),$$

where the integer coefficients are the entries in Tables 9 and 10.

By construction, $\mathcal{L}$ is invariant under the symmetry group $\mathcal{G}$ of the configuration: each orbit average is unchanged by the action of $\mathcal{G}$. It is also invariant under interchanging

red and blue, since this fixes g , sends s to $-s$, and the expression contains s only through products $s(v)s(w)$.

Lemma 22. *Let $V = \{0, 1, 2\}^2$, and define*

$$\mathcal{E}_1 = \{\{p, q\} \subseteq V : Q(p - q) = 1\},$$

$$\mathcal{I}_3 = \{(p, p + u, p + 2u) : u \in \{(1, 0), (0, 1), (1, -1)\}, p, p + u, p + 2u \in V\}.$$

Let $\mathcal{A}_3$ be the set of colorings $c : V \rightarrow \{G, R, B\}$ for which no pair in $\mathcal{E}_1$ has color RR or BB , and no triple in $\mathcal{I}_3$ has color GGG . Then $|\mathcal{A}_3| = 304$, and $4\mathcal{L}(c) \geq 0$ for every $c \in \mathcal{A}_3$, with equality for at least one coloring.

Proof. The set $\mathcal{E}_1$ contains the 16 unit pairs in the configuration, and $\mathcal{I}_3$ contains the 7 copies of ℓ_3 in the three unit directions $(1, 0)$, $(0, 1)$, and $(1, -1)$. For a coloring satisfying these local restrictions, put $g_p = 1_{\{c(p)=G\}}$ and $s_p = 1_{\{c(p)=R\}} - 1_{\{c(p)=B\}}$. Then

$$\begin{aligned} F(c) = 4\mathcal{L}(c) = & -784 + \sum_{O \in \mathcal{O}_V} \frac{4c_O}{|O|} \sum_{p \in O} g_p \\ & + \sum_{U \in \mathcal{O}_P} \frac{4}{|U|} \left(a_U \sum_{\{p, q\} \in U} g_p g_q + b_U \sum_{\{p, q\} \in U} s_p s_q \right). \end{aligned}$$

The orbit sizes are 1, 2, or 4, so $4/|O|$ and $4/|U|$ are integers. also $g_p, s_p \in \{-1, 0, 1\}$. Thus $4\mathcal{L}(c) = F(c)$ is integer-valued. There are 304 colorings satisfying the local restrictions, and substitution gives $F(c) \geq 0$ for each of them. $\square$

Lemma 23. *If a coloring of Λ contains no red ℓ_2 , no blue ℓ_2 , and no green ℓ_3 , then $\mathcal{L} \geq 0$.*

Proof. The condition that no red or blue unit pair occurs is the condition that no pair in $\mathcal{E}_1$ has color RR or BB , because $Q(p - q) = 1$ lists every unit pair of Λ . Likewise, the stated definition of $\mathcal{I}_3$ lists every three-point unit-spaced collinear copy of ℓ_3 in Λ . Hence the coloring belongs to $\mathcal{A}_3$, and the preceding lemma gives $4\mathcal{L} \geq 0$. $\square$

4.2. Averaging the local expression. Suppose that a measurable coloring of $\mathbb{R}^2$ contains none of the three configurations in Theorem 2, and apply the same definitions of g and s pointwise. For $x \in \mathbb{R}^2$ and $\theta \in \mathbb{R}$, let $\text{Col}_{x, \theta}$ be the coloring induced on $x + \rho_\theta \Lambda$, and set

$$A_T = \frac{1}{2\pi|B_T|} \int_0^{2\pi} \int_{B_T} \mathcal{L}(\text{Col}_{x, \theta}) dx d\theta.$$

Lemma 23 gives $A_T \geq 0$.

Grouping the coefficients by squared distance gives

$$\begin{aligned} H_g(t) = & -603 - 1050J_0(t) + 1610J_0(\sqrt{3}t) - 1152J_0(2t) \\ & + 1000J_0(\sqrt{7}t) + 375J_0(\sqrt{12}t), \end{aligned} \tag{4.1}$$

$$\begin{aligned} H_s(t) = & -442J_0(t) + 138J_0(\sqrt{3}t) - 12J_0(2t) \\ & + 300J_0(\sqrt{7}t) + 200J_0(\sqrt{12}t). \end{aligned} \tag{4.2}$$

Proposition 24. *For every $T > 2\sqrt{3}$,*

$$\left| A_T + 196 - \int_0^\infty H_g(t) d\nu_{g,T}(t) - \int_0^\infty H_s(t) d\nu_{s,T}(t) \right| \leq \frac{134272\sqrt{3}}{T}. \quad (4.3)$$

Moreover,

$$\nu_{g,T}([0, \infty)) + \nu_{s,T}([0, \infty)) = 1. \quad (4.4)$$

Proof. Every point of Λ has norm at most $\sqrt{3}$. The absolute sums of the linear and quadratic coefficients are 2599 and 10323, respectively, so Proposition 15 gives the error

$$\frac{4\sqrt{3}}{T} 2599 + \frac{12\sqrt{3}}{T} 10323 = \frac{134272\sqrt{3}}{T}.$$

The vertex coefficients sum to -603 , and grouping the coefficients for pairs by distance gives (4.1)–(4.2). Hence (4.3). Pointwise $g + s^2 = 1$, so Lemma 11 gives (4.4). $\square$

4.3. Bounds for the kernels.

Proposition 25. *For every $t \geq 0$, one has $H_g(t) < 193$ and $H_s(t) < 193$.*

Proof. On $[0, 25]$, Appendix A gives $H_g(t_j) \leq 180$ and $H_s(t_j) < 46019/250$ at $t_j = j/400$. Lemma 17 and the chain rule give $\|H'_g\|_\infty < 30452/3$ and $\|H'_s\|_\infty < 4415/2$ on this interval. The mean value theorem then gives

$$H_g(t) < 180 + \frac{30452/3}{800} < 193, \quad H_s(t) < \frac{46019}{250} + \frac{4415/2}{800} < 193.$$

For $t \geq 25$, Lemma 17 and (4.1)–(4.2), using $3^{-1/4} < 10/13$, $4^{-1/4} < 5/7$, $7^{-1/4} < 5/8$, and $12^{-1/4} < 5/9$, give $H_g(t) < 55$ and $H_s(t) < 143$. $\square$

Proof of Theorem 2. Choose $T = 100000$. The measures $\nu_{g,T}$ and $\nu_{s,T}$ are nonnegative, so Propositions 24 and 25, together with (4.4), give the following chain. This is another application of Proposition 16, with the nine-point local inequality providing the lower bound and the two kernel estimates providing the upper bound.

$$\begin{aligned} 0 &\leq A_T \\ &\leq -196 + \int_0^\infty H_g d\nu_{g,T} + \int_0^\infty H_s d\nu_{s,T} + \frac{134272\sqrt{3}}{T} \\ &\leq -196 + 193(\nu_{g,T}([0, \infty)) + \nu_{s,T}([0, \infty))) + \frac{134272\sqrt{3}}{T} \\ &= -3 + \frac{134272\sqrt{3}}{T} \\ &< 0. \end{aligned}$$

Since $134272\sqrt{3} < 300000$, the final inequality holds for the chosen T . This contradiction proves Theorem 2. $\square$

5. PROOFS OF THEOREMS 3–6

5.1. Line inequalities and averaging. The four proofs begin with the same finite directed-graph lemma.

Lemma 26. *Let $G = (V, E)$ be a finite directed graph, assign a real weight $w : E \rightarrow \mathbb{R}$ to each edge, and let $\phi : V \rightarrow \mathbb{R}$ satisfy*

$$w(u, v) + \phi(u) - \phi(v) \geq 0$$

on every edge, then every directed path $v_0, \dots, v_N$ satisfies

$$\sum_{j=0}^{N-1} w(v_j, v_{j+1}) \geq \phi(v_N) - \phi(v_0).$$

Proof. For a directed path $v_0, \dots, v_N$, summing the edge inequalities gives

$$\begin{aligned} 0 &\leq \sum_{j=0}^{N-1} (w(v_j, v_{j+1}) + \phi(v_j) - \phi(v_{j+1})) \\ &= \sum_{j=0}^{N-1} w(v_j, v_{j+1}) + \sum_{j=0}^{N-1} (\phi(v_j) - \phi(v_{j+1})) \\ &= \sum_{j=0}^{N-1} w(v_j, v_{j+1}) + \phi(v_0) - \phi(v_N). \end{aligned}$$

Rearranging gives the stated inequality. $\square$

The function ϕ records the contribution of the current local pattern. When the local inequality is summed along a line, all intermediate values cancel and only the two endpoint values remain. The averaging argument below is the same cancellation with a boundary error.

For a red–blue coloring, let $r = 1_R$ and $s = 1_R - 1_B$. Let $\omega \in \mathbb{S}^{d-1}$ be a unit direction vector, so $x + \mathbb{R}\omega$ is the line through x in direction ω . On this oriented line, write $r_j = r(x + j\omega)$ and $s_j = s(x + j\omega)$. If there is no red ℓ_a and no blue ℓ_b , then the binary sequence $(r_j)_{j \in \mathbb{Z}}$ contains neither the consecutive string 1^a nor the consecutive string 0^b . In the $\{-1, 1\}$ -encoding, the absence of a monochromatic ℓ_4 says that no four consecutive signs are equal.

The graph vertices below are represented by binary sequences. Thus, in the (ℓ_4, ℓ_4) case, put

$$\epsilon_j = r_j = \frac{s_j + 1}{2} \in \{0, 1\}.$$

The sequence $\epsilon_0 \cdots \epsilon_3$ avoids the forbidden consecutive patterns unless it is 0000 or 1111, while the signed variables in W_{44} are obtained from the same sequence by $s_j = 2\epsilon_j - 1$. In the other three cases the entries are already the binary values used by r_j .

Let $Q = \{34, 35, 36, 44\}$. For $q \in Q$, define F_q and m_q by

q	F_q	m_q
34	$\{111, 0000\}$	3
35	$\{111, 00000\}$	6
36	$\{111, 000000\}$	6
44	$\{1111, 0000\}$	4

A finite binary sequence avoids F_q when it contains no member of F_q consecutively. Define the vertex and edge sets by

$$\mathcal{V}_q = \{u \in \{0, 1\}^{m_q} : u \text{ avoids } F_q\},$$

and

$$\mathcal{E}_q = \left\{ (u, v) \in \mathcal{V}_q^2 : \begin{array}{l} u_1 \cdots u_{m_q-1} = v_0 \cdots v_{m_q-2}, \\ u_0 \cdots u_{m_q-1} v_{m_q-1} \text{ avoids } F_q \end{array} \right\}.$$

Thus forbidden consecutive patterns are excluded both from the vertex set and from the length- $(m_q + 1)$ sequence associated with an edge. The direction $u \rightarrow v$ moves one position to the left and appends the next color, so it records adding one point along the oriented line.

The sequence length is chosen so that the forbidden patterns and every entry in the local expression are determined by one edge. A coloring of an oriented line therefore gives a directed path in the corresponding graph. The resulting numbers of vertices and edges are recorded in Table 1 in the appendix. For an allowed edge, define

$$\begin{aligned} W_{34} &= 4r_0 - 1 - 3r_0r_1 - 2r_0r_2, \\ W_{35} &= -6 + 28r_0 - 20r_0r_1 - 14r_0r_2 - 3r_0r_3 - 4r_0r_4 + 2r_0r_5 + 2r_0r_6, \\ W_{36} &= -2 + 11r_0 - 9r_0r_1 - 5r_0r_2 - 2r_0r_4 - r_0r_5 + r_0r_6, \\ W_{44} &= -1 - 6s_0s_1 - 4s_0s_2 - 2s_0s_3 + s_0s_4. \end{aligned}$$

The symbols r_j and s_j denote the colors at the successive points of the length- $(m_q + 1)$ sequence associated with the edge. The expressions are chosen so that their averages over all translated and oriented lines involve only the correlations listed below. The four expressions and accompanying vertex potentials are the integer data recorded in Appendix B. The four potentials are listed there explicitly for every vertex in the corresponding set $\mathcal{V}_q$. Denote them by $\phi_q : \mathcal{V}_q \rightarrow \mathbb{Z}$. Only these listed values are used below. The exhaustive verification gives the following common form of the four edge inequalities.

Proposition 27. *For every $q \in Q$ and every allowed edge $(u, v) \in \mathcal{E}_q$,*

$$W_q(u, v) + \phi_q(u) - \phi_q(v) \geq 0.$$

Proof. For each q , set

$$\sigma_q(u, v) = W_q(u, v) + \phi_q(u) - \phi_q(v).$$

For example, in the (ℓ_3, ℓ_4) graph the edge $000 \rightarrow 001$ is associated with the sequence 0001, so

$$\begin{aligned} W_{34}(0001) &= -1, \\ \sigma_{34}(000, 001) &= -1 + \phi_{34}(000) - \phi_{34}(001) \\ &= -1 + 0 - (-1) = 0. \end{aligned}$$

Substituting the integer data from Appendix B gives $\sigma_q(u, v) \geq 0$ for every allowed edge in each of the four graphs. $\square$

Define

$$\begin{aligned} L_{34,T} &= 4p_T(r) - 1 - 3C_{1,T}(r) - 2C_{2,T}(r), \\ L_{35,T} &= -6 + 28p_T(r) - 20C_{1,T}(r) - 14C_{2,T}(r) - 3C_{3,T}(r) \\ &\quad - 4C_{4,T}(r) + 2C_{5,T}(r) + 2C_{6,T}(r), \\ L_{36,T} &= -2 + 11p_T(r) - 9C_{1,T}(r) - 5C_{2,T}(r) - 2C_{4,T}(r) - C_{5,T}(r) + C_{6,T}(r), \\ L_{44,T} &= -1 - 6C_{1,T}(s) - 4C_{2,T}(s) - 2C_{3,T}(s) + C_{4,T}(s). \end{aligned}$$

To control the endpoint terms, for each $q \in Q$ put $M_q = \max_{u \in \mathcal{V}_q} |\phi_q(u)|$. Extend the potential from the finite vertex set to all binary sequences of length m_q by

$$\bar{\phi}_q(z) = \begin{cases} \phi_q(z), & z \in \mathcal{V}_q, \\ 0, & z \notin \mathcal{V}_q, \end{cases}$$

where $z \in \{0, 1\}^{m_q}$, and then to $\mathbb{R}^{m_q}$ by

$$\Phi_q(y_0, \dots, y_{m_q-1}) = \bar{\phi}_q(1_{\{y_0 > 1/2\}}, \dots, 1_{\{y_{m_q-1} > 1/2\}}).$$

Thus Φ_q is bounded and measurable on $\mathbb{R}^{m_q}$, with $|\Phi_q| \leq M_q$. In particular, it satisfies the boundedness hypothesis of Lemma 12 on the relevant cube.

Proposition 28. *If the corresponding two monochromatic lines are absent, then, for $T \geq 1$,*

$$L_{q,T} \geq -\frac{M_q D_{d_q}}{T}.$$

Proof. Fix q and (x, ω) . In all four cases let $z_j = r(x + j\omega)$. In the (ℓ_4, ℓ_4) case this is the binary sequence $z_j = \epsilon_j$, and the signed variables in W_{44} are recovered from $s_j = 2z_j - 1$. The avoidance assumption implies that every length- m_q sequence encountered along the line belongs to $\mathcal{V}_q$, and that every length- $(m_q + 1)$ sequence associated with an edge contains no forbidden pattern. Hence the successive vertices form a directed path in the graph. Apply Proposition 27 to the edge corresponding to the length- $(m_q + 1)$ sequence beginning at x , and average this inequality over $B_T \times \mathbb{S}^{d_q-1}$.

The normalized averages of r_0 , $r_0 r_k$, and $s_0 s_k$ are, by definition, $p_T(r)$, $C_{k,T}(r)$, and $C_{k,T}(s)$. The remaining two terms are the averages of

$$\Phi_q(z_0(x, \omega), \dots, z_{m_q-1}(x, \omega)) - \Phi_q(z_1(x, \omega), \dots, z_{m_q}(x, \omega)).$$

The shift by one point along the oriented line is the source of this boundary term. The second part of Lemma 12, applied to the bounded extension Φ_q , bounds its absolute value by $M_q D_{d_q}/T$. Removing it leaves the stated lower bound. $\square$

For the next proposition, fix a finite nonempty set K of positive integers, real numbers a, b , and real coefficients $(c_k)_{k \in K}$. Let $r = 1_R$ be a measurable red indicator, write $p = p_T(r)$ and $f_T = r - p_T(r)$, and set

$$S := \sum_{k \in K} c_k,$$

$$H(t) := \sum_{k \in K} c_k \Omega_d(kt).$$

Define

$$L_{r,T} := a + bp + \sum_{k \in K} c_k C_{k,T}(r).$$

Let $\mu_{f,T}$ be the centered-indicator measure associated with f_T in Lemma 14. For a signed coloring $s = 1_R - 1_B$, define

$$L_{s,T} := a + \sum_{k \in K} c_k C_{k,T}(s)$$

and let $\nu_{s,T}$ be the measure associated with s_T from the same lemma. Both measures are nonnegative, with masses $p(1-p)$ and 1, respectively.

Proposition 29. *For $T \geq \max K$, there are errors $E_{r,T}$ and $E_{s,T}$ such that*

$$L_{r,T} = a + bp + Sp^2 + \int_0^\infty H(t) d\mu_{f,T}(t) + E_{r,T}, \quad |E_{r,T}| \leq \frac{2D_d}{T} \sum_{k \in K} |c_k|k,$$

$$L_{s,T} = a + \int_0^\infty H(t) d\nu_{s,T}(t) + E_{s,T}, \quad |E_{s,T}| \leq \frac{D_d}{T} \sum_{k \in K} |c_k|k.$$

Proof. By Lemma 14, for each $k \in K$ there is an error $E_k(T)$ such that, for $T \geq \max K$,

$$C_{k,T}(r) = p^2 + \int_0^\infty \Omega_d(kt) d\mu_{f,T}(t) + E_k(T),$$

$$|E_k(T)| \leq \frac{2D_d k}{T}.$$

Multiplying by c_k and summing over K , using the definitions of $L_{r,T}$, S , and H , gives

$$L_{r,T} = a + bp + Sp^2 + \int_0^\infty H(t) d\mu_{f,T}(t) + \sum_{k \in K} c_k E_k(T).$$

Set $E_{r,T} = \sum_{k \in K} c_k E_k(T)$. The triangle inequality gives

$$|E_{r,T}| \leq \frac{2D_d}{T} \sum_{k \in K} |c_k|k.$$

If r is replaced by $s = 1_R - 1_B$, the second formula in Lemma 14 gives, for each $k \in K$,

$$C_{k,T}(s) = \int_0^\infty \Omega_d(kt) d\nu_{s,T}(t) + \tilde{E}_k(T),$$

$$|\tilde{E}_k(T)| \leq \frac{D_d k}{T},$$

and summing after multiplication by c_k yields

$$L_{s,T} = a + \int_0^\infty H(t) d\nu_{s,T}(t) + \sum_{k \in K} c_k \tilde{E}_k(T),$$

where $E_{s,T} = \sum_{k \in K} c_k \tilde{E}_k(T)$. The triangle inequality gives the stated bound. $\square$

For the next proposition, retain the notation of Proposition 29. Suppose a finite directed graph has vertices u , edges $u \rightarrow v$, and a function ϕ on its vertices. Assume that its edge inequalities have the form $W(u, v) + \phi(u) - \phi(v) \geq 0$, so that summing along a line gives, for every coloring avoiding the target configurations, one of the bounds $L_{r,T} \geq -\frac{\kappa_{\text{pot}}}{T}$ or $L_{s,T} \geq -\frac{\kappa_{\text{pot}}}{T}$ for some $\kappa_{\text{pot}} \geq 0$. Let $B \in \mathbb{R}$ satisfy $H(t) \leq B$ for all $t \geq 0$.

In the red-indicator case, assume that

$$a + bp + Sp^2 + Bp(1 - p) \leq -\eta \quad (0 \leq p \leq 1)$$

for some $\eta > 0$. In the signed-function case, assume instead that $a + B \leq -\eta$. Define the averaging-error constant

$$\kappa_{\text{avg}} = \begin{cases} 2D_d \sum_{k \in K} |c_k|k, & \text{in the red-indicator case,} \\ D_d \sum_{k \in K} |c_k|k, & \text{in the signed-function case.} \end{cases}$$

Proposition 30. *Under these assumptions, the corresponding measurable coloring cannot exist. The contradiction follows for*

$$T > \max \left\{ \max K, \frac{\kappa_{\text{pot}} + \kappa_{\text{avg}}}{\eta} \right\}.$$

Proof. In the red-indicator case, Proposition 29, the positivity of $\mu_{f,T}$, and its mass $p(1 - p)$ give

$$L_{r,T} \leq a + bp + Sp^2 + Bp(1 - p) + \frac{\kappa_{\text{avg}}}{T} \leq -\eta + \frac{\kappa_{\text{avg}}}{T}.$$

The signed-function case is the same calculation with the mass-one measure $\nu_{s,T}$:

$$L_{s,T} \leq a + B + \frac{\kappa_{\text{avg}}}{T} \leq -\eta + \frac{\kappa_{\text{avg}}}{T}.$$

For the stated range of T , either upper bound is smaller than the lower bound supplied by the directed-graph argument. $\square$

For the four coefficient vectors, put

$$\begin{aligned} H_{34}(t) &= -3\Omega_3(t) - 2\Omega_3(2t), \\ H_{35}(t) &= -20\Omega_4(t) - 14\Omega_4(2t) - 3\Omega_4(3t) - 4\Omega_4(4t) + 2\Omega_4(5t) + 2\Omega_4(6t), \\ H_{36}(t) &= -9\Omega_7(t) - 5\Omega_7(2t) - 2\Omega_7(4t) - \Omega_7(5t) + \Omega_7(6t), \\ H_{44}(t) &= -6\Omega_4(t) - 4\Omega_4(2t) - 2\Omega_4(3t) + \Omega_4(4t). \end{aligned}$$

Proposition 31. *For every $t \geq 0$,*

$$\begin{aligned} H_{34}(t) &< \frac{7}{10}, & H_{35}(t) &< \frac{59}{20}, \\ H_{36}(t) &< \frac{2}{5}, & H_{44}(t) &< \frac{9}{10}. \end{aligned}$$

Proof. Table 4 gives the sampled values, spacings, and derivative bounds on $[0, 6]$, $[0, 25]$, $[0, 10]$, and $[0, 25]$, respectively. The mean value theorem gives

$$\begin{aligned} H_{34}(t) &< \frac{69}{100} + \frac{7}{2} \frac{1}{10000} < \frac{7}{10}, & H_{35}(t) &< \frac{1469}{500} + \frac{95}{2} \frac{1}{4000} < \frac{59}{20}, \\ H_{36}(t) &< \frac{399}{1000} + \frac{95}{8} \frac{1}{20000} < \frac{2}{5}, & H_{44}(t) &< \frac{887}{1000} + 12 \frac{1}{5000} < \frac{9}{10}. \end{aligned}$$

On the complementary ranges, Lemma 17 applied termwise gives

$$\begin{aligned} H_{34}(t) &\leq 4/t \leq 2/3 & (t \geq 6), \\ H_{35}(t) &\leq \frac{2}{t} \left(20 + \frac{14}{2} + \frac{3}{3} + \frac{4}{4} + \frac{2}{5} + \frac{2}{6} \right) < \frac{59}{20} & (t \geq 25), \\ H_{36}(t) &\leq \frac{30}{t^3} \left(9 + \frac{5}{2^3} + \frac{2}{4^3} + \frac{1}{5^3} + \frac{1}{6^3} \right) < \frac{2}{5} & (t \geq 10), \\ H_{44}(t) &\leq \frac{2}{t} \left(6 + \frac{4}{2} + \frac{2}{3} + \frac{1}{4} \right) < \frac{9}{10} & (t \geq 25). \end{aligned}$$

□

For the quantitative estimates below, the integer potentials in Proposition 27 have the bounds on their absolute values in the M_q column of the following table. The last two columns record the boundary constants supplied by Propositions 28 and 29, respectively.

case	D_{d_q}	M_q	$\sum_k c_{q,k} k$	κ_q^{pot}	κ_q^{avg}
34	24	3	7	72	336
35	64	30	95	1920	12160
36	896	13	38	11648	68096
44	64	14	24	896	1536

The constants κ_q^{pot} and κ_q^{avg} come from the potential argument and the averaging estimate, respectively. We have $\kappa_q^{\text{pot}} = M_q D_{d_q}$. For the first three rows $\kappa_q^{\text{avg}} = 2D_{d_q} \sum_k |c_{q,k}|k$, while for the last row the factor 2 is absent because the coloring is encoded by s .

To make the last step explicit, write $p = p_T(r)$. In the first three cases, Proposition 29 and positivity of $\mu_{f,T}$ allow the integral of H_{ij} to be replaced by its upper bound times $p(1-p)$. The sums of the correlation coefficients are -5 , -37 , and -16 , respectively.

Substitution of the bounds from Proposition 31 and completion of the square give

$$L_{34,T} \leq -\frac{57}{10} \left(p_T - \frac{47}{114} \right)^2 - \frac{71}{2280} + \frac{336}{T}, \quad (5.1)$$

$$L_{35,T} \leq -\frac{799}{20} \left(p_T - \frac{619}{1598} \right)^2 - \frac{359}{63920} + \frac{12160}{T}, \quad (5.2)$$

$$L_{36,T} \leq -\frac{82}{5} \left(p_T - \frac{57}{164} \right)^2 - \frac{31}{1640} + \frac{68096}{T}, \quad (5.3)$$

$$L_{44,T} \leq -\frac{1}{10} + \frac{1536}{T}. \quad (5.4)$$

For the last line, the measure associated with s has mass 1, so $L_{44,T} \leq -1 + 9/10 + 1536/T$.

The next four proofs apply Proposition 30. The edge inequalities give the lower bounds, while (5.1)–(5.4) give the corresponding Fourier upper bounds.

5.2. Proof of Theorem 3.

Proof. Assume for contradiction that a measurable red–blue coloring of $\mathbb{E}^3$ contains no red ℓ_3 and no blue ℓ_4 . The lower and upper bounds are

$$-\frac{72}{T} \leq L_{34,T} \leq -\frac{71}{2280} + \frac{336}{T}.$$

They would force

$$T \leq \frac{(72 + 336) \cdot 2280}{71} = \frac{930240}{71},$$

contradicting a choice of T above this threshold. This proves Theorem 3. $\square$

5.3. Proof of Theorem 4.

Proof. Assume for contradiction that a measurable red–blue coloring of $\mathbb{E}^4$ contains no red ℓ_3 and no blue ℓ_5 . The lower and upper bounds are

$$-\frac{1920}{T} \leq L_{35,T} \leq -\frac{359}{63920} + \frac{12160}{T}.$$

They would force

$$T \leq \frac{(1920 + 12160) \cdot 63920}{359} = \frac{899993600}{359},$$

contradicting a choice of T above this threshold. This proves Theorem 4. $\square$

5.4. Proof of Theorem 5.

Proof. Assume for contradiction that a measurable red–blue coloring of $\mathbb{E}^7$ contains no red ℓ_3 and no blue ℓ_6 . The lower and upper bounds are

$$-\frac{11648}{T} \leq L_{36,T} \leq -\frac{31}{1640} + \frac{68096}{T}.$$

They would force

$$T \leq \frac{(11648 + 68096) \cdot 1640}{31} = \frac{130780160}{31},$$

contradicting a choice of T above this threshold. This proves Theorem 5. $\square$

5.5. Proof of Theorem 6.

Proof. Assume for contradiction that a measurable red–blue coloring of $\mathbb{E}^4$ contains no red ℓ_4 and no blue ℓ_4 . The lower and upper bounds are

$$-\frac{896}{T} \leq L_{44,T} \leq -\frac{1}{10} + \frac{1536}{T}.$$

They would force

$$T \leq (896 + 1536) \cdot 10 = 24320,$$

contradicting a choice of T above this threshold. This proves Theorem 6. $\square$

The following two planar proofs use Proposition 16. In each case the finite local lemma gives the lower bound for the averaged expression, and the kernel estimate gives the upper bound.

6. PROOF OF THEOREM 7

The configurations used in the two planar polygon arguments are shown in Figure 3. The first is the union of four centered copies of $W_4(1)$. The second is the configuration used for P_5 .

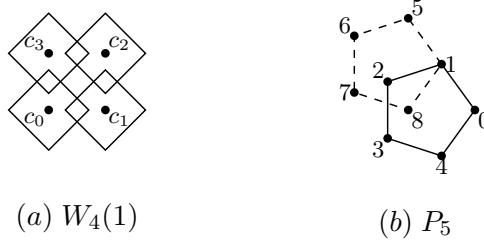


FIGURE 3. The planar configurations used for $W_4(1)$ and P_5 .

6.1. The finite square–center inequality. Put $R = 1/\sqrt{2}$ and let

$$U_R = \{0, Re_1, Re_2, -Re_1, -Re_2\}.$$

Place four translated copies of U_R at the vertices of the unit square. Label the points in the four copies by

$$v_{5k} = c_k, \quad v_{5k+1} = c_k + Re_1, \quad v_{5k+2} = c_k + Re_2, \quad v_{5k+3} = c_k - Re_1, \quad v_{5k+4} = c_k - Re_2,$$

where $c_0 = (0, 0)$, $c_1 = (1, 0)$, $c_2 = (1, 1)$, and $c_3 = (0, 1)$. The four copies of $W_4(1)$ have index sets

$$I_k = \{5k, 5k + 1, 5k + 2, 5k + 3, 5k + 4\} \quad (0 \leq k \leq 3).$$

The unit-distance pairs among these points are

$$\begin{aligned} \mathcal{E}_4 = \{ & (0, 5), (0, 15), (1, 2), (1, 4), (1, 6), (1, 16), \\ & (2, 3), (2, 7), (2, 17), (3, 4), (3, 8), (3, 18), \\ & (4, 9), (4, 19), (5, 10), (6, 7), (6, 9), (6, 11), \\ & (7, 8), (7, 12), (8, 9), (8, 13), (9, 14), (10, 15), \\ & (11, 12), (11, 14), (11, 16), (12, 13), (12, 17), (13, 14), \\ & (13, 18), (14, 19), (16, 17), (16, 19), (17, 18), (18, 19)\}. \end{aligned}$$

The notation (i, j) denotes the unordered pair $\{i, j\}$.

Write $x_i = 1$ when v_i is red and $x_i = 0$ when it is blue. A coloring with no red unit pair and no blue copy of $W_4(1)$ must satisfy

$$\begin{aligned} x_i x_j &= 0 \text{ for } (i, j) \in \mathcal{E}_4, \\ \sum_{i \in I_k} x_i &\geq 1 \quad (0 \leq k \leq 3). \end{aligned}$$

The first condition excludes red unit pairs, and the last four conditions say that each displayed square-center copy contains a red point.

For $x \in \{0, 1\}^{20}$, define

$$\mathcal{F}_{W_4(1)}(x) = c + \sum_i \alpha_i x_i + \sum_{i < j} \beta_{ij} x_i x_j,$$

where the coefficient data, written in the common 10^6 -scaled form described in Appendix B, are listed in Tables 6 and 7.

Lemma 32. *If no pair in $\mathcal{E}_4$ is simultaneously red and each of the four sets I_k contains a red point, then $\mathcal{F}_{W_4(1)} \geq 0$.*

Proof. The finite set of binary colorings satisfying the local restrictions is

$$\mathcal{A} = \left\{ x \in \{0, 1\}^{20} : \begin{array}{l} x_i x_j = 0 \text{ for every } (i, j) \in \mathcal{E}_4, \\ \sum_{i \in I_k} x_i \geq 1 \text{ for } 0 \leq k \leq 3 \end{array} \right\}.$$

Enumerating the 2^{20} binary colorings gives $|\mathcal{A}| = 2338$. Evaluating the shifted quadratic expression on this set gives

$$\min_{x \in \mathcal{A}} \mathcal{F}_{W_4(1)}(x) = 0.$$

Thus $\mathcal{F}_{W_4(1)}(x) \geq 0$ for every such vector. □

6.2. The averaged kernel. The corresponding radial kernel is

$$K(t) = \sum_{i < j} \beta_{ij} J_0(|v_i - v_j|t).$$

The distance-grouped coefficients are recorded in Table 8 in the appendix.

Proposition 33. *For every $t \geq 0$,*

$$K(t) - 600J_0(t) < 46.$$

Proof. Appendix A gives interval bounds for $K(t) - 600J_0(t)$ on $[0, 125]$, and the standard Bessel tail estimate gives $|K(t) - 600J_0(t)| < 46$ for $t \geq 125$. The two estimates prove the proposition. $\square$

Proof of Theorem 7. Assume that a measurable coloring has no red unit pair and no blue copy of $W_4(1)$. Let r be the red indicator. Along a sequence of growing balls, write $p_T = p_T(r)$ and choose a subsequence for which $p_T \rightarrow p$. For every x and θ , each of the four displayed copies contains a red point, so $\sum_{i=0}^{19} r(x + \rho_\theta v_i) \geq 4$. The translation and rotation averages of the summands tend to p , and hence $20p \geq 4$, so $p \geq 1/5$. The measurable unit-distance density bound gives $p < 1/4$ [2].

The local lemma gives $A_T \geq 0$. Proposition 15 gives an error $E_T = o(1)$, since the configuration has radius less than 2. The mass of the centered Fourier measure is $p_T(1 - p_T)$, and the no-red-unit-pair condition gives

$$\int_0^\infty J_0(t) d\mu_{f,T}(t) = -p_T^2 + o(1).$$

Using Proposition 33 and passing to the limit, the density calculation in Appendix B gives

$$\limsup_{T \rightarrow \infty} A_T < 0$$

for $p \in [1/5, 1/4]$, contradicting $A_T \geq 0$. This proves the theorem. $\square$

7. PROOF OF THEOREM 8

7.1. The finite pentagon inequality. The two overlapping copies of P_5 used in this proof are shown in Figure 3.

Put

$$R_5 = \frac{1}{2 \sin(\pi/5)}, \quad u_j = R_5 \left(\cos \frac{2\pi j}{5}, \sin \frac{2\pi j}{5} \right), \quad s = u_1 - u_0.$$

The two copies of P_5 have vertex sets

$$B_0 = \{u_0, u_1, u_2, u_3, u_4\}, \quad B_1 = \{u_0 + s, u_1 + s, u_2 + s, u_3 + s, u_4 + s\}.$$

After identifying the common point $u_0 + s = u_1$, label the nine distinct points by $v_0, \dots, v_8$ in the order

$$v_i = u_i \quad (0 \leq i \leq 4), \quad v_{4+j} = u_j + s \quad (1 \leq j \leq 4).$$

The identity $u_0 + s = u_1$ gives the common point. There is no other common point: a chord of this regular pentagon of length 1 is a side, and the ten directed side vectors are distinct, so $u_i - u_j = u_1 - u_0$ forces $(i, j) = (1, 0)$ (with indices taken modulo 5). Within the two pentagons, the side pairs are $(0, 1), (1, 2), (2, 3), (3, 4), (4, 0)$ and $(1, 5), (5, 6), (6, 7), (7, 8), (8, 1)$. For completeness, the calculation for pairs joining the two pentagons uses

$$R_5^2 = \frac{5 + \sqrt{5}}{10}, \quad |u_i - (u_j + s)|^2 = |u_i - u_j - u_1 + u_0|^2,$$

and

$$\cos 0 = 1, \quad \cos \frac{2\pi}{5} = \frac{\sqrt{5} - 1}{4}, \quad \cos \frac{4\pi}{5} = -\frac{\sqrt{5} + 1}{4}.$$

For $i \in \{0, 1, 2, 3, 4\}$ and $j \in \{1, 2, 3, 4\}$, the unit pairs joining the two pentagons are

$$(i, j) = (1, 1), (1, 4), (2, 2), (3, 3), (4, 4).$$

The first two are already the translated-pentagon sides $(1, 5)$ and $(1, 8)$. The other three give $(2, 6), (3, 7), (4, 8)$. Thus the complete unit-pair list is

$$\mathcal{E}_5 = \{(0, 1), (0, 4), (1, 2), (1, 5), (1, 8), (2, 3), (2, 6), (3, 4), (3, 7), (4, 8), (5, 6), (6, 7), (7, 8)\}.$$

The corresponding index sets are

$$I_0 = \{0, 1, 2, 3, 4\}, \quad I_1 = \{1, 5, 6, 7, 8\}.$$

Write $x_i = 1$ when v_i is red and $x_i = 0$ when it is blue. A coloring with no red unit pair and no blue copy of P_5 must satisfy

$$\begin{aligned} x_i x_j &= 0 \text{ for } (i, j) \in \mathcal{E}_5, \\ x_0 + x_1 + x_2 + x_3 + x_4 &\geq 1, \\ x_1 + x_5 + x_6 + x_7 + x_8 &\geq 1. \end{aligned}$$

The first condition excludes red unit pairs. The last two conditions say that each pentagon contains a red point. A pentagon lacking a red point would be an all-blue copy of P_5 .

For $x \in \{0, 1\}^9$, define

$$\mathcal{F}_{P_5}(x) = -77 + \sum_i \alpha_i x_i + \sum_{i < j} \beta_{ij} x_i x_j,$$

where the coefficient vector and the upper-triangular matrix (β_{ij}) are listed in Table 15 in the appendix.

Lemma 34. *If no pair in $\mathcal{E}_5$ is simultaneously red and each pentagon contains a red point, then $\mathcal{F}_{P_5} \geq 0$.*

Proof. For $x = (x_0, \dots, x_8) \in \{0, 1\}^9$, consider the vectors for which $x_i x_j = 0$ for $(i, j) \in \mathcal{E}_5$, $\sum_{i=0}^4 x_i \geq 1$, and $x_1 + x_5 + x_6 + x_7 + x_8 \geq 1$. There are 42 such colorings, and substitution into the polynomial gives $\mathcal{F}_{P_5}(x) \geq 0$ for each of them. Before the shift the minimum was 1. $\square$

7.2. The pentagon kernel. There are 36 quadratic pairs, but eight have zero coefficient. Computing the squared distances of the remaining 28 pairs in $\mathbb{Q}(\sqrt{5})$ and grouping their coefficients gives

squared distance	number of nonzero-coefficient pairs	coefficient sum
$\frac{3 - \sqrt{5}}{2}$	3	200
$\frac{1}{2}$	13	-1048
$\frac{5 - \sqrt{5}}{2}$	2	100
$\frac{3 + \sqrt{5}}{2}$	10	-385.

Every nonzero quadratic coefficient occurs in one of these four classes. The eight zero coefficients contribute nothing. Hence these are the only squared distances with nonzero coefficient sum, and all other distance classes disappear from the kernel. Consequently,

$$\begin{aligned} K_5(t) = & 200J_0\left(\sqrt{\frac{3-\sqrt{5}}{2}}t\right) - 1048J_0(t) \\ & + 100J_0\left(\sqrt{\frac{5-\sqrt{5}}{2}}t\right) - 385J_0\left(\sqrt{\frac{3+\sqrt{5}}{2}}t\right). \end{aligned}$$

Proposition 35. *For every $t \geq 0$, $K_5(t) < 393$.*

Proof. On $[0, 12]$, Appendix A gives $K_5(\tau_k) < 389$ at the midpoints of a grid with spacing $1/200$, and $|K'_5| < 15949/12$. Hence the mean value theorem gives

$$K_5(t) < 389 + \frac{15949}{4800} < 393.$$

For $t \geq 12$, put $q_1 = (3 - \sqrt{5})/2$, $q_3 = (5 - \sqrt{5})/2$, and $q_4 = (3 + \sqrt{5})/2$. Lemma 17 gives $|K_5(t)| \leq \sqrt{2/(\pi t)} \sum_i |c_i| q_i^{-1/4}$ for the four terms in K_5 , with $q_2 = 1$. The bounds

$$q_1^{1/4} > \frac{39}{50}, \quad q_3^{1/4} > \frac{27}{25}, \quad q_4^{1/4} > \frac{127}{100}$$

give $\sum_i |c_i| q_i^{-1/4} < 1701$. Since $\pi > 3.14$ and $t \geq 12$, $\sqrt{2/(\pi t)} < 5/\sqrt{471}$, and therefore $|K_5(t)| < 392 < 393$. $\square$

Proof of Theorem 8. Assume that a measurable coloring has no red unit pair and no blue copy of P_5 . Let r be the red indicator, put $p = p_T(r)$, and set $f_T = r - p_T(r)$. The local lemma gives $A_T \geq 0$. Proposition 15 gives

$$A_T = -77 + 285p - 1133p^2 + \int_0^\infty K_5(t) d\mu_{f,T}(t) + E_T,$$

where $\mu_{f,T}$ has mass $p(1-p)$. Since $\sum_i |\alpha_i| = 623$, $\sum_{i < j} |\beta_{ij}| = 2163$, and the finite configuration has radius less than 2, Proposition 15 gives $|E_T| \leq 56896/T$. Using $K_5 < 393$,

$$A_T \leq -77 + 678p - 1526p^2 + \frac{56896}{T} \leq -\frac{2581}{1526} + \frac{56896}{T},$$

where the last inequality follows by completing the square. At $T = 200000$ the right side is negative, contradicting $A_T \geq 0$. $\square$

8. PROOF OF THEOREM 9

8.1. The three-cube inequality. Put $e = (1, 0, 0)$ and define

$$\begin{aligned} v_0 &= (0, 1, 0), & w_0 &= (0, 0, 1), \\ v_1 &= (0, 1/2, \sqrt{3}/2), & w_1 &= (0, -\sqrt{3}/2, 1/2), \\ v_2 &= (0, -1/2, \sqrt{3}/2), & w_2 &= (0, -\sqrt{3}/2, -1/2). \end{aligned}$$

For $r \in \{0, 1, 2\}$, put $C_r = \{ae + bv_r + cw_r : a, b, c \in \{0, 1\}\}$ and let $\mathcal{P} = C_0 \cup C_1 \cup C_2$. The definitions give $e \cdot v_r = e \cdot w_r = v_r \cdot w_r = 0$ and $|e| = |v_r| = |w_r| = 1$. Thus, for each r , the vectors e, v_r, w_r form an orthonormal basis, and C_r is the vertex set of a unit cube. The three cubes share the edge $\{0, e\}$, and their union $\mathcal{P}$ has twenty points. For $Q_r = \{bv_r + cw_r : b, c \in \{0, 1\}\}$, the three cross-sections Q_r perpendicular to the common edge are shown in Figure 4; the full configuration $\mathcal{P}$ is obtained by translating each Q_r by 0 and by e .

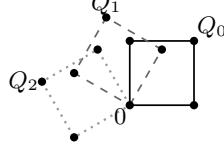


FIGURE 4. The three cross-sections Q_r of the configuration $\mathcal{P}$ in the plane perpendicular to the common edge e . Solid, dashed, and dotted lines represent Q_0 , Q_1 , and Q_2 , respectively.

Consider the Euclidean isometries

$$\tau(x, y, z) = (1 - x, y, z)$$

and

$$\sigma(x, y, z) = \left(x, -\frac{\sqrt{3}}{2}y - \frac{1}{2}z, -\frac{1}{2}y + \frac{\sqrt{3}}{2}z \right),$$

and write $G = \langle \tau, \sigma \rangle$.

Label the points by

$$\begin{aligned} p_0 &= 0, & p_1 &= w_0, & p_2 &= v_0, & p_3 &= v_0 + w_0, \\ p_4 &= e, & p_5 &= e + w_0, & p_6 &= e + v_0, & p_7 &= e + v_0 + w_0, \\ p_8 &= w_1, & p_9 &= v_1, & p_{10} &= v_1 + w_1, & p_{11} &= e + w_1, \\ p_{12} &= e + v_1, & p_{13} &= e + v_1 + w_1, & p_{14} &= w_2, & p_{15} &= v_2, \\ p_{16} &= v_2 + w_2, & p_{17} &= e + w_2, & p_{18} &= e + v_2, & p_{19} &= e + v_2 + w_2. \end{aligned}$$

The following classification is needed because the six pairwise distances alone allow more than one configuration in three dimensions. For example, the points p_0, p_1, p_4, p_8 have four unit pairs and two pairs at distance $\sqrt{2}$, but

$$\det[p_1 - p_0, p_4 - p_0, p_8 - p_0] = -\frac{\sqrt{3}}{2} \neq 0.$$

They are noncoplanar and form a copy of $\mathcal{T}$.

Lemma 36. *Let $X \subset \mathbb{E}^3$ have four pairs at distance 1 and two pairs at distance $\sqrt{2}$. If the two pairs at distance $\sqrt{2}$ are disjoint, then X is a unit square. If they share a vertex, then X is isometric to $\mathcal{T}$.*

Proof. If the two $\sqrt{2}$ -pairs are disjoint, label them $\{x_1, x_3\}$ and $\{x_2, x_4\}$. After translating the midpoint of x_1x_3 to the origin, $x_3 = -x_1$ and $|x_1|^2 = 1/2$. Since each of x_2, x_4 is at distance 1 from both x_1 and x_3 , we have $x_1 \perp x_2, x_4$ and $|x_2|^2 = |x_4|^2 = 1/2$. The

relation $|x_2 - x_4|^2 = 2$ then gives $x_4 = -x_2$. Thus $X = \{\pm x_1, \pm x_2\}$ with $x_1 \perp x_2$ and $|x_1| = |x_2| = 1/\sqrt{2}$, so X is a unit square.

If the pairs share a vertex, label the points $0, a, b, c$ so that $|a - b| = |b - c| = \sqrt{2}$. The other four distances are 1, so the Gram matrix of (a, b, c) is

$$\begin{pmatrix} 1 & 0 & 1/2 \\ 0 & 1 & 0 \\ 1/2 & 0 & 1 \end{pmatrix},$$

which is also the Gram matrix of $e_1, e_2, \frac{1}{2}e_1 + \frac{\sqrt{3}}{2}e_3$. A Gram matrix records all inner products after the common base point has been fixed, so two such lists of vectors are related by an orthogonal map on their spans. That map extends to an orthogonal map of $\mathbb{E}^3$. Thus Gram matrices determine the configuration up to Euclidean isometry, and X is isometric to $\mathcal{T}$. $\square$

On the labels, the generators act as

$$\begin{aligned} \sigma &= (1\ 15)(2\ 14)(3\ 16)(5\ 18)(6\ 17)(7\ 19)(8\ 9)(11\ 12), \\ \tau &= (0\ 4)(1\ 5)(2\ 6)(3\ 7)(8\ 11)(9\ 12)(10\ 13)(14\ 17)(15\ 18)(16\ 19). \end{aligned}$$

Lemma 37. *The set $\mathcal{P}$ has twenty points. Among its four-point subsets there are twenty-two copies of $\mathcal{S}$ and twenty-four copies of $\mathcal{T}$. Moreover, G has order four, preserves $\mathcal{P}$, and has fifty-six orbits on the unordered pairs of points of $\mathcal{P}$.*

Proof. All coordinates lie in $\mathbb{Q}(\sqrt{3})$. Appendix A.3 records the six squared distances for each of the $\binom{20}{4}$ four-point subsets. It identifies forty-six subsets whose six squared distances, counted with multiplicity, are $\{1, 1, 1, 1, 2, 2\}$. Among these, the two pairs at squared distance 2 are disjoint in twenty-two cases and share a vertex in twenty-four cases. Lemma 36 therefore identifies these cases as twenty-two copies of $\mathcal{S}$ and twenty-four copies of $\mathcal{T}$.

For $a, b, c \in \{0, 1\}$, the formulas defining τ and σ are

$$\tau(ae + bv_r + cw_r) = (1 - a)e + bv_r + cw_r,$$

while $\sigma e = e$, $(\sigma v_0, \sigma w_0) = (w_2, v_2)$, $(\sigma v_1, \sigma w_1) = (w_1, v_1)$, and $(\sigma v_2, \sigma w_2) = (w_0, v_0)$. Thus τ preserves each C_r , while σ exchanges C_0 and C_2 and preserves C_1 . Both maps preserve $\mathcal{P}$. They are commuting Euclidean involutions, neither is the identity, and $\tau \neq \sigma$ since τ changes the first coordinate whereas σ fixes it. Thus G has order four. The induced action on the $\binom{20}{2} = 190$ unordered pairs has 2 orbits of size 1, 14 orbits of size 2, and 40 orbits of size 4. Hence there are 56 pair orbits, accounting for $2 \cdot 1 + 14 \cdot 2 + 40 \cdot 4 = 190$ pairs. $\square$

Index the pair orbits by $\mathcal{O}_0, \dots, \mathcal{O}_{55}$ in the order of the representatives in Appendix B. This row order is the definition of the indices. For a coloring encoded by $s : \mathcal{P} \rightarrow \{-1, 1\}$, define $X_j(s) = \sum_{\{p_u, p_v\} \in \mathcal{O}_j} s(p_u)s(p_v)$. Let a_j be the corresponding integer coefficient. Thus $X_j(s)$ is the sum of the products $s(p_u)s(p_v)$ over the pairs in the orbit in row j of

Appendix B, and a_j is the coefficient in that same row. The symmetry reduction gives

$$\mathcal{L}_S(s) = \frac{1}{1000} \left(-36608 + \sum_{j=0}^{55} a_j X_j(s) \right). \quad (8.1)$$

The four-point subsets in the local verification are those whose six squared distances, counted with multiplicity, are $\{1, 1, 1, 1, 2, 2\}$. Both a unit square and $\mathcal{T}$ have these six squared distances, and Lemma 36 says that every subset with these distances is one of these two configurations. Hence the forty-six forbidden subsets are the twenty-two copies of $\mathcal{S}$ and the twenty-four copies of $\mathcal{T}$ from Lemma 37.

Lemma 38. *If $s : \mathcal{P} \rightarrow \{-1, 1\}$ contains neither a monochromatic copy of $\mathcal{S}$ nor a monochromatic copy of $\mathcal{T}$, then $\mathcal{L}_S(s) \geq 0$.*

Proof. Let $R = \{p \in \mathcal{P} : s(p) = 1\}$ be the red set. Among the 2^{20} subsets $R \subseteq \mathcal{P}$, keep those for which no one of the forty-six forbidden four-point subsets is contained in R or in $\mathcal{P} \setminus R$. There are 47946 such subsets. Substitution of the integer coefficients from Appendix B gives the shifted numerator at least 0 in each case, and hence $\mathcal{L}_S(s) \geq 0$. $\square$

We now average the local inequality. Suppose that a measurable red–blue coloring of $\mathbb{R}^3$ contains neither a monochromatic unit square nor a monochromatic copy of $\mathcal{T}$, and put $s = 1_R - 1_B : \mathbb{R}^3 \rightarrow \{-1, 1\}$. For $x \in \mathbb{R}^3$, $U \in \text{SO}(3)$, and $p \in \mathcal{P}$, put $s_{x,U}(p) = s(x + Up)$, and define

$$A_T = \frac{1}{|B_T|} \int_{\text{SO}(3)} \int_{B_T} \mathcal{L}_S(s_{x,U}) \, dx \, dU,$$

where dU is the probability measure on $\text{SO}(3)$ invariant under rotations. Lemma 38 gives

$$A_T \geq 0. \quad (8.2)$$

For each squared distance q occurring in $\mathcal{P}$, define

$$B_q = \sum_{\substack{0 \leq j \leq 55 \\ |p_u - p_v|^2 = q \text{ on } \mathcal{O}_j}} |\mathcal{O}_j| a_j.$$

The coefficient sums B_q are listed in Table 20 in the appendix. In particular, $\sum_q |B_q| = 620268$. Define

$$K_S(t) = \frac{1}{1000} \left(-36608 + \sum_q B_q \Omega_3(\sqrt{q}t) \right). \quad (8.3)$$

Proposition 39. *There is a nonnegative measure $\nu_{s,T}$ on radial Fourier variables of mass 1 such that*

$$A_T = \int_0^\infty K_S(t) \, d\nu_{s,T}(t) + O(T^{-1}).$$

For $T > 2\sqrt{3}$,

$$\left| A_T - \int_0^\infty K_S(t) \, d\nu_{s,T}(t) \right| \leq \frac{301383\sqrt{3}}{50T}. \quad (8.4)$$

Proof. Take $\nu_{s,T}$ from Lemma 11. For a fixed pair $p_u, p_v \in \mathcal{P}$, the substitution $y = x + Up_u$ gives

$$\int_{B_T} s(x + Up_u)s(x + Up_v) dx = \int_{B_T + Up_u} s(y)s(y + U(p_v - p_u)) dy.$$

Let $A = B_T + Up_u$ and $C = B_T \cap (B_T - U(p_v - p_u))$. Since $|p_u|, |p_v| \leq \sqrt{3}$, both A and C contain $B_{T-2\sqrt{3}}$ and are contained in $B_{T+\sqrt{3}}$. In other words,

$$B_{T-2\sqrt{3}} \subseteq A \cap C \subseteq B_{T+\sqrt{3}}.$$

Thus

$$\frac{|A \triangle C|}{|B_T|} \leq \frac{(T + \sqrt{3})^3 - (T - 2\sqrt{3})^3}{T^3} \leq \frac{9\sqrt{3}}{T}.$$

The difference between the normalized integrals is therefore at most $9\sqrt{3}/T$. Averaging over U and applying Lemma 11 gives

$$\frac{1}{|B_T|} \int_{\text{SO}(3)} \int_{B_T} s(x + Up_u)s(x + Up_v) dx dU = \int_0^\infty \Omega_3(|p_u - p_v|t) d\nu_{s,T}(t) + O(T^{-1}).$$

Each pair in the orbit $\mathcal{O}_j$ carries the same coefficient a_j . Therefore the sum of the absolute coefficients over individual pairs is

$$\sum_{j=0}^{55} \sum_{\{p_u, p_v\} \in \mathcal{O}_j} |a_j| = \sum_{j=0}^{55} |\mathcal{O}_j| |a_j|.$$

From the coefficient data,

$$\sum_{j=0}^{55} |\mathcal{O}_j| |a_j| = 669740.$$

Consequently, after the factor $1/1000$ in (8.1), the total boundary error is at most $9\sqrt{3} \cdot 669740/(1000T) = 301383\sqrt{3}/(50T)$. Substituting the pair identities into (8.1) and grouping terms by the fourteen squared distances gives (8.3). Finally, $s^2 = 1$ pointwise, so

$$\nu_{s,T}([0, \infty)) = \frac{1}{|B_T|} \int_{B_T} s(x)^2 dx = 1.$$

□

8.2. Bounds for the kernel.

Proposition 40. *For every $t \geq 0$, one has $K_S(t) < -209/400$.*

Proof. Appendix A.3 gives the interval bound

$$K_S(j/1000) < -\frac{247}{250} \tag{8.5}$$

on the grid with spacing $1/1000$ in $[0, 40]$. Every squared distance in (8.3) is less than 9. Lemma 10 gives $|\Omega'_3| \leq 1/2$, so the chain rule gives $|K'_S(t)| \leq (1/2000) \sum_q |B_q| \sqrt{q} < 931$.

If $0 \leq t \leq 40$, choose j with $|t - j/1000| \leq 1/2000$. The mean value theorem and (8.5) give

$$K_S(t) < -\frac{247}{250} + \frac{931}{2000} = -\frac{209}{400}.$$

Now suppose $t \geq 40$. The smallest squared distance in the kernel is $2 - \sqrt{3} > 1/4$, so every corresponding distance is greater than $1/2$ and every argument $\sqrt{q}t$ is greater than $20 > 2$. Thus the large-argument estimate $|\Omega_3(u)| \leq 1/u$ from Lemma 17 applies and gives

$$K_S(t) \leq -\frac{36608}{1000} + \frac{2 \cdot 620268}{1000t} \leq -\frac{27973}{5000} < -\frac{209}{400}.$$

□

Proof of Theorem 9. Suppose that a measurable red–blue coloring of $\mathbb{R}^3$ contains neither a monochromatic unit square nor a monochromatic copy of $\mathcal{T}$. Equation (8.2) gives $A_T \geq 0$. On the other hand, Proposition 39, the positivity and unit mass of $\nu_{s,T}$, and Proposition 40 give

$$A_T \leq -\frac{209}{400} + \frac{301383\sqrt{3}}{50T}.$$

The upper bound is negative whenever

$$\frac{301383\sqrt{3}}{50T} < \frac{209}{400} \iff T > \frac{301383 \cdot 400}{50 \cdot 209} \sqrt{3} = \frac{2411064}{209} \sqrt{3}.$$

This contradiction proves Theorem 9. □

9. LINEAR PROGRAMS

The finite inequalities above were found with linear programs. We record the formulations because they make the role of the coefficients transparent. The LPs are used to choose real coefficients. After rounding, the proofs use the resulting rational or integer data. The local inequalities are then verified on the finite sets of colorings and sequences, while the bounds for all $t \geq 0$ follow from the analytic estimates in the proof sections. The objective in each LP is to maximize a gap δ between the local lower bound and the averaged upper bound.

We count scalar inequalities and equalities separately. A two-sided bound on a variable contributes two inequalities. For each LP, let $\mathcal{N}$ denote the finite set of kernel sampling conditions used in its construction. In the planar quadratic LPs an element of $\mathcal{N}$ may include the density parameter as well as the frequency. The sampling sets for the main kernel constraints are

$$\begin{aligned} \mathcal{N}_7 &= \{j/400 : 0 \leq j \leq 40000\}, & \mathcal{N}_3 &= \{j/400 : 0 \leq j \leq 10000\}, \\ \mathcal{N}_{34} &= \{j/5000 : 0 \leq j \leq 30000\}, & \mathcal{N}_{35} &= \{j/2000 : 0 \leq j \leq 50000\}, \\ \mathcal{N}_{36} &= \{j/10000 : 0 \leq j \leq 100000\}, & \mathcal{N}_{44} &= \{j/2500 : 0 \leq j \leq 62500\}, \\ \mathcal{N}_{S,T} &= \{j/1000 : 0 \leq j \leq 40000\}. \end{aligned}$$

For P_5 , write $\mathcal{N}_{P_5}$ for the finite set of sampled (p, t) pairs used in its quadratic LP. The $W_4(1)$ coefficients are verified by the separate four-copy program described in the proof. The cardinalities, together with the dimensions and scalar constraint counts of the displayed LPs, are recorded in Appendix A. The sampled constraints are used only to select the coefficients. The proofs use the rounded data and the separate analytic verifications below.

9.1. Planar quadratic inequalities. We first describe the common planar setup. Let V be a finite point set, let $B_1, \dots, B_s \subseteq V$ be the local copies of the target configuration, and let E be the set of forbidden unit pairs. Write $r_v \in \{0, 1\}$ for the indicator that v is red. The allowed local colorings are the vectors $r = (r_v)_{v \in V}$ satisfying

$$r_u r_v = 0 \text{ for } \{u, v\} \in E, \sum_{v \in B_i} r_v \geq 1 \text{ for } 1 \leq i \leq s.$$

The first condition excludes a red unit pair, and the second says that each local copy has a red point. For LP coefficients c , a_v , and $b_{uv} = b_{vu}$, consider

$$\tilde{\mathcal{L}}(r) = c + \sum_{v \in V} a_v r_v + \sum_{\{u, v\} \subseteq V} b_{uv} r_u r_v.$$

For a measurable global red indicator, write $p = p_T(r)$ and $f_T = r - p_T(r)$. For $0 < p < 1$, normalize the measure on radial Fourier variables associated with the centered function f_T by its mass $p(1 - p)$. At a fixed frequency t , the contribution of the term $r_u r_v$ in the rotationally averaged correlation is

$$p^2 + p(1 - p)J_0(|u - v|t).$$

The full average integrates this expression against the normalized nonnegative measure on radial Fourier variables. The average depends on the full measure. A single frequency is insufficient to determine it. If $p = 0$ or $p = 1$, the centered measure is zero and the pair contribution is simply p^2 . Since the measure is nonnegative, pointwise bounds in p and t are enough. The function J_0 is the Bessel function of order zero. Thus the averaged expression is

$$\tilde{H}(p, t) = c + p \sum_{v \in V} a_v + \sum_{\{u, v\} \subseteq V} b_{uv} (p^2 + p(1 - p)J_0(|u - v|t)).$$

The corresponding LP has variables $\delta, c, (a_v)_{v \in V}, (b_{uv})_{\{u, v\} \subseteq V}$ and objective maximize δ , subject to

$$\begin{aligned} &\text{maximize} && \delta \\ &\text{variables} && \delta, c, (a_v)_{v \in V}, (b_{uv})_{\{u, v\} \subseteq V} \\ &\text{subject to} && \tilde{\mathcal{L}}(r) \geq 0 \text{ for every allowed coloring } r, \\ & && \tilde{H}(p, t) \leq -\delta \text{ for every sampled pair } (p, t), \\ & && 0 \leq \delta \leq 100, \\ & && |c|, |a_v|, |b_{uv}| \leq 100. \end{aligned}$$

The coefficient of each variable is fixed by the local geometry and the chosen samples. For P_5 , the point set has 9 points, so its quadratic LP has $2 + 9 + \binom{9}{2} = 47$ variables. Its number of allowed colorings and kernel inequalities is given in Table 5. The $W_4(1)$ calculation uses the separate 20-point four-copy verification described above.

The eighteen-point and nine-point planar inequalities use the same principle, with symmetry and color indicators reducing the number of variables. For the eighteen-point configuration, let N_R and N_B count red and blue points, and let $\mathcal{A}_7$ be the set of colorings with no red unit pair and no blue copy of ℓ_7 . For variables x_v and $y_{uv} = y_{vu}$,

define

$$\begin{aligned}\tilde{\mathcal{L}}_7(r) &= -1001 + \frac{500}{9}N_R + \frac{500}{9}N_B + \sum_v x_v r_v + \sum_{\{u,v\}} y_{uv} r_u r_v, \\ \tilde{H}_R(t) &= 1000 + \sum_v x_v + \sum_{\{u,v\}} y_{uv} J_0(|u-v|t), \\ \tilde{H}_B(t) &= 1000,\end{aligned}$$

The two indicator measures make the averaging behind this formulation explicit. If p is the red density, then, apart from the boundary error,

$$\tilde{A}_{7,T} = -1001 + \int_0^\infty \tilde{H}_R(u) d\nu_{r,T}(u) + \int_0^\infty \tilde{H}_B(u) d\nu_{b,T}(u) + O(T^{-1}).$$

The corresponding centered (p, t) expression, after the constant linear terms cancel, is

$$\tilde{\mathcal{H}}_7(p, t) = -1 + p \sum_v x_v + p^2 \sum_{\{u,v\}} y_{uv} + p(1-p) \sum_{\{u,v\}} y_{uv} J_0(|u-v|t).$$

The two measures in the preceding average have masses p and $1-p$. The LP below optimizes only the red kernel $\tilde{H}_R$ against the fixed reference value $\tilde{H}_B = 1000$. This is the pointwise kernel bound used in the proof, while the expression above records its relation to the full average.

The LP is

$$\begin{aligned}\text{maximize} \quad & \delta \\ \text{variables} \quad & \delta, (x_v)_{v \in P}, (y_{uv})_{\{u,v\} \subseteq P} \\ \text{subject to} \quad & \tilde{\mathcal{L}}_7(r) \geq 0 \text{ for } r \in \mathcal{A}_7, \\ & \tilde{H}_R(t) \leq 1000 - \delta \text{ for } t \in \mathcal{N}_7.\end{aligned}$$

This LP has $172 = 1 + 18 + 153$ variables and 751 local inequalities. The number of kernel inequalities is given in Table 5. The remaining range $t \geq 100$ is covered by the Bessel estimate in the proof of the eighteen-point kernel proposition.

The nine-point three-color LP has variables δ , four vertex-orbit coefficients, and thirteen coefficients for each of two types of pair sums, for a total of $31 = 1 + 4 + 13 + 13$ variables. If $\mathcal{A}_3$ denotes the 304 colorings satisfying its local conditions, its local expression is

$$\tilde{\mathcal{L}}_3 = -196 + \sum_O x_O [g]_O + \sum_U (y_U [gg]_U + z_U [ss]_U),$$

with averaged expressions

$$\begin{aligned}\tilde{H}_g(t) &= \sum_O x_O + \sum_U y_U J_0(\sqrt{d(U)}t), \\ \tilde{H}_s(t) &= \sum_U z_U J_0(\sqrt{d(U)}t).\end{aligned}$$

In terms of the two measures, the averaged expression is

$$\tilde{A}_{3,T} = -196 + \int_0^\infty \tilde{H}_g(t) d\nu_{g,T}(t) + \int_0^\infty \tilde{H}_s(t) d\nu_{s,T}(t) + O(T^{-1}),$$

and the two nonnegative measures have masses whose sum is 1. Thus $\widetilde{H}_g$ and $\widetilde{H}_s$ are the two averaged kernels, including the constant term in $\widetilde{H}_g$. The LP optimizes their common pointwise upper bound. The density parameter is handled separately.

The LP is

$$\begin{aligned} & \text{maximize} && \delta \\ & \text{variables} && \delta, (x_O)_O, (y_U)_U, (z_U)_U \\ & \text{subject to} && \widetilde{\mathcal{L}}_3(r) \geq 0 \text{ for } r \in \mathcal{A}_3, \\ & && \widetilde{H}_g(t) \leq 196 - \delta \text{ for } t \in \mathcal{N}_3, \\ & && \widetilde{H}_s(t) \leq 196 - \delta \text{ for } t \in \mathcal{N}_3. \end{aligned}$$

The LP has $31 = 1 + 4 + 13 + 13$ variables and 304 local inequalities. The number of kernel inequalities is given in Table 5. The common bound for $\widetilde{H}_g$ and $\widetilde{H}_s$ is needed because the two associated measures have total mass 1.

9.2. Linear programs for the line graphs. For the four higher-dimensional line results, the finite object is a directed graph of allowed local sequences. Let $\mathcal{T}_q$ be its directed edge set, with initial and terminal vertices $u(z)$ and $v(z)$. The LP variables are the coefficients in a local expression $\widetilde{W}_q$, the potential values $\widetilde{\phi}_q(u)$ on the vertices, and the gap δ_q . The local constraints are

$$\widetilde{W}_q(z) + \widetilde{\phi}_q(u(z)) - \widetilde{\phi}_q(v(z)) \geq 0 \text{ for } z \in \mathcal{T}_q.$$

The four local expressions used in the LPs are

$$\begin{aligned} \widetilde{W}_{34}(r) &= -1 + a_{34}r_0 + c_{34,1}r_0r_1 + c_{34,2}r_0r_2, \\ \widetilde{W}_{35}(r) &= -6 + a_{35}r_0 + \sum_{k=1}^6 c_{35,k}r_0r_k, \\ \widetilde{W}_{36}(r) &= -2 + a_{36}r_0 + \sum_{k \in \{1,2,4,5,6\}} c_{36,k}r_0r_k, \\ \widetilde{W}_{44}(s) &= -1 + \sum_{k=1}^4 c_{44,k}s_0s_k. \end{aligned}$$

Writing Ω_d for the spherical averaging function in dimension d , put $p = p_T(r)$ and $f_T = r - p_T(r)$. The associated kernel expressions are $\widetilde{H}_q(t) = \sum_k c_{q,k} \Omega_{d_q}(kt)$. The fixed local constants and the linear terms are handled separately from $\widetilde{H}_q$. For $q = 34, 35, 36$, the full averaged expression has the form

$$\widetilde{\mathcal{L}}_{q,T} = \alpha_q + \beta_q p + S_q p^2 + \int_0^\infty \widetilde{H}_q(t) d\mu_{f,T}(t) + O(T^{-1}),$$

where

q	α_q	β_q	S_q
34	-1	4	-5
35	-6	28	-37
36	-2	11	-16

and $\mu_{f,T}$ has mass $p(1-p)$. For $q = 44$, the signed-function identity is

$$\tilde{\mathcal{L}}_{44,T} = -1 + \int_0^\infty \tilde{H}_{44}(t) d\nu_{s,T}(t) + O(T^{-1}), \text{ and } \nu_{s,T}([0, \infty)) = 1.$$

Thus these line LPs optimize only the pointwise kernel bounds $\tilde{H}_q(t) \leq B_q - \delta_q$ for $t \in \mathcal{N}_q$. The polynomials in the color density in the first display are handled separately by completing the square in the theorem proofs. The LP optimizes the kernel bounds, while the theorem proofs handle those polynomials in p . The sample sets and upper bounds are

q	d_q	B_q
34	3	7/10
35	4	59/20
36	7	2/5
44	4	9/10

For definiteness, impose the bounds

$$\begin{aligned} -100 &\leq \delta_q \leq 100, \\ |a_q|, |c_{q,k}| &\leq 100, \\ |\tilde{\phi}_q(u)| &\leq 1000. \end{aligned}$$

The variable a_q is omitted when $q = 44$. These bounds make the LP bounded. The proofs use only the rounded coefficients and potentials.

For each row q , choose a vertex u_q^* and impose the normalization $\tilde{\phi}_q(u_q^*) = 0$. The four LPs can then be written together as

$$\begin{aligned} &\text{maximize} && \delta_q \\ &\text{variables} && \delta_q, a_q, (c_{q,k})_k, (\tilde{\phi}_q(u))_u \\ &\text{subject to} && \tilde{W}_q(z) + \tilde{\phi}_q(u(z)) - \tilde{\phi}_q(v(z)) \geq 0 \text{ for } z \in \mathcal{T}_q, \\ & && \tilde{H}_q(t) \leq B_q - \delta_q \text{ for } t \in \mathcal{N}_q, \\ & && \tilde{\phi}_q(u_q^*) = 0, \\ & && -100 \leq \delta_q \leq 100, \\ & && |a_q|, |c_{q,k}| \leq 100, \\ & && |\tilde{\phi}_q(u)| \leq 1000. \end{aligned}$$

In the (ℓ_4, ℓ_4) row, the variable a_q is omitted. The numbers of variables and all scalar constraint counts are collected in Table 5. The sample sets $\mathcal{N}_q$ are specified above.

The rounded variable coefficients are

q	a_q	$(c_{q,k})_k$
34	4	$(-3, -2)$
35	28	$(-20, -14, -3, -4, 2, 2)$
36	11	$(-9, -5, -2, -1, 1)$ for $k = 1, 2, 4, 5, 6$
44	—	$(-6, -4, -2, 1)$.

The corresponding integer-valued functions ϕ_q complete the four rounded inequalities. The local and kernel verifications use the rounded data throughout.

9.3. The square–tetrahedron LP. The square–tetrahedron LP uses signs $s_v \in \{-1, 1\}$ on the twenty-point configuration and groups the 56 pair terms into orbits under the configuration symmetries. For each orbit, let $X_j(s)$ be the sum of the products of the signs on its pairs. The local expression and its averaged kernel are

$$\begin{aligned}\tilde{\mathcal{L}}_{\mathcal{S}, \mathcal{T}}(s) &= c + \sum_{j=0}^{55} b_j X_j(s), \\ \tilde{K}_{\mathcal{S}, \mathcal{T}}(t) &= c + \sum_{j=0}^{55} |\mathcal{O}_j| b_j \Omega_3(\sqrt{q_j} t).\end{aligned}$$

Let $\mathcal{A}_{\mathcal{S}, \mathcal{T}}$ be the set of the 47946 colorings by signs $s : \mathcal{P} \rightarrow \{-1, 1\}$ that avoid both monochromatic configurations. The LP is

$$\begin{aligned}\text{maximize} \quad & \delta \\ \text{variables} \quad & \delta, c, (b_j)_{0 \leq j \leq 55} \\ \text{subject to} \quad & \tilde{\mathcal{L}}_{\mathcal{S}, \mathcal{T}}(s) \geq 0 \text{ for } s \in \mathcal{A}_{\mathcal{S}, \mathcal{T}}, \\ & \tilde{K}_{\mathcal{S}, \mathcal{T}}(t) \leq -\delta \text{ for } t \in \mathcal{N}_{\mathcal{S}, \mathcal{T}}, \\ & |c| \leq 50 \text{ and } |b_j| \leq 50 \text{ for } 0 \leq j \leq 55.\end{aligned}$$

These are the colorings that avoid the twenty-two four-point subsets forming squares and the twenty-four four-point subsets forming copies of $\mathcal{T}$, as classified by the three-cube geometry lemma.

It has $58 = 1 + 1 + 56$ variables. There are 47946 allowed colorings by signs, and the dimensions and scalar constraint counts are given in Table 5. The bounds on c and the b_j only make the LP bounded. They play no role in the mathematical statement.

10. FURTHER CONSEQUENCES AND OPEN PROBLEMS

10.1. Further consequences. The same local inequalities also give finite-ball versions. For $R > 0$, let $B_R^d = \{x \in \mathbb{E}^d : |x| \leq R\}$. For finite configurations $X_1, \dots, X_r \subset \mathbb{E}^d$, write $B_R^d \rightarrow_m (X_1, \dots, X_r)$ if every measurable coloring $B_R^d = C_1 \sqcup \dots \sqcup C_r$ contains, for some i , an isometric copy of X_i all of whose points lie in C_i .

Corollary 41. *The following seven finite-ball relations hold with the indicated radii:*

$$\begin{aligned}B_R^2 &\rightarrow_m (\ell_2, \ell_7) && \text{if } R > \frac{109469}{2} \sqrt{73}, \\ B_R^2 &\rightarrow_m (\ell_2, \ell_2, \ell_3) && \text{if } R > \frac{134275}{3} \sqrt{3}, \\ B_R^3 &\rightarrow_m (\ell_3, \ell_4) && \text{if } R > \frac{930453}{71}, \\ B_R^4 &\rightarrow_m (\ell_3, \ell_5) && \text{if } R > \frac{899995754}{359}, \\ B_R^7 &\rightarrow_m (\ell_3, \ell_6) && \text{if } R > \frac{130780346}{31}, \\ B_R^4 &\rightarrow_m (\ell_4, \ell_4) && \text{if } R > 24324, \\ B_R^2 &\rightarrow_m (\ell_2, P_5) && \text{if } R > 200002.\end{aligned}$$

Moreover, if $R > \frac{2411273}{209} \sqrt{3}$, every measurable two-coloring of B_R^3 contains a monochromatic copy of $\mathcal{S}$ or a monochromatic copy of $\mathcal{T}$.

Proof. Repeat the corresponding averaging proof over B_T^d , extending the color indicators by zero outside B_R^d . It is enough that every translated local configuration used in the average remain inside B_R^d . Thus we may take $T = R - \rho$, where, in order, the required

values of ρ are $\frac{\sqrt{73}}{2}$, $\sqrt{3}$, 3, 6, 6, 4, 2, and $\sqrt{3}$. The first two values are the radii of the planar configurations. The next four are the largest shifts along the local line. The polygon configuration has radius at most 2. Every point of the square–tetrahedron configuration has norm at most $\sqrt{3}$.

The corresponding arguments give contradictions whenever

$$\begin{aligned} T &> 54734\sqrt{73}, & T &> \frac{134272}{3}\sqrt{3}, & T &> \frac{930240}{71}, \\ T &> \frac{899993600}{359}, & T &> \frac{130780160}{31}, & T &> 24320, \\ T &> 200000, & T &> \frac{2411064}{209}\sqrt{3}. \end{aligned}$$

Adding the corresponding value of ρ gives the stated radii. $\square$

Corollary 42. *For each relation with a finite-radius bound above, every measurable coloring has a color class containing infinitely many pairwise disjoint isometric copies of the corresponding configuration. Moreover, every measurable two-coloring of $\mathbb{E}^3$ contains infinitely many pairwise disjoint monochromatic copies of at least one of $\mathcal{S}$ and $\mathcal{T}$.*

Proof. For one of these relations, choose infinitely many pairwise disjoint balls with a radius satisfying its finite-radius bound. Restrict the coloring to each ball and apply Corollary 41, after translating the ball to B_R^d . Each ball then contains one of the finitely many color–configuration pairs (i, X_i) . By the pigeonhole principle, one such pair occurs in infinitely many balls. The corresponding copies are pairwise disjoint because they lie in pairwise disjoint balls.

For the square–tetrahedron result, use balls with the corresponding finite radius. Each ball contains a monochromatic copy of $\mathcal{S}$ or of $\mathcal{T}$. There are only four choices of color and configuration, so one choice occurs in infinitely many balls. The corresponding copies are pairwise disjoint. $\square$

10.2. Open problems. The results suggest three broad questions.

Problem 1. *Can the measurability assumption be removed from the results of this paper? Equivalently, do all Ramsey relations proved here hold for arbitrary colorings? In particular, does the relation in Theorem 1 hold with no measurability assumption?*

Problem 2. *Can the gaps between the known positive and negative results be narrowed? In particular:*

- (i) *does $\mathbb{E}^2 \rightarrow_m (\ell_2, \ell_8)$ hold?*
- (ii) *can any of the dimensions in Theorems 3–6 be lowered?*
- (iii) *does $\mathbb{E}^3 \rightarrow_m (\mathcal{S}, \mathcal{S})$ hold?*
- (iv) *does $\mathbb{E}^2 \rightarrow_m (\ell_2, P_7)$ hold?*

Problem 3. *Can the finite-ball relations in Corollary 41 be proved with smaller radii?*

APPENDIX A. COMPUTATIONAL VERIFICATIONS

The code used for the computational verifications is available in the GitHub repository.

The coefficient data are recorded in Appendix B. The local inequalities are verified by enumeration. On bounded frequency intervals, computer programs evaluate the relevant series with rational arithmetic (planar and line cases) or interval arithmetic with outward rounding (square–tetrahedron case). Derivative bounds extend the sampled values between sample points.

A.1. Local computational verifications. The local verifications use only the rational or integer data in Appendix B. The numerical solutions of the LPs are not used below. For the eighteen-point configuration, enumerate the red subsets $R \subseteq P$ satisfying the local restrictions and evaluate the integer polynomial

$$F(R) = -1 + \sum_{p \in P} c_p r_p(R) + \sum_{\{p,q\} \subset P} a_{pq} r_p(R) r_q(R),$$

where the shifted constant is the remaining term after the color-count terms combine. There are 751 allowed subsets, and evaluating $F(R)$ gives $F(R) \geq 0$ for every one of them.

For the nine-point three-color configuration, the local restrictions leave 304 colorings. With $g = 1_G$ and $s = 1_R - 1_B$, evaluating the integer expression $F = 4\mathcal{L}$ from the preceding lemma gives $F \geq 0$.

For the four line graphs, the allowed binary sequences have lengths 3, 6, 6, 4. Their forbidden consecutive patterns and the numbers of vertices and edges are as follows. Join

case	forbidden consecutive patterns	vertices	edges
(ℓ_3, ℓ_4)	111, 0000	7	12
(ℓ_3, ℓ_5)	111, 00000	41	73
(ℓ_3, ℓ_6)	111, 000000	43	78
(ℓ_4, ℓ_4)	1111, 0000	14	26

TABLE 1. Sizes of the four finite line graphs.

two sequences when their overlapping entries agree and the resulting extended sequence contains no forbidden pattern. Substituting the integer potentials in Appendix B gives $W_q + \phi_q(u) - \phi_q(v) \geq 0$ for every resulting edge.

For $W_4(1)$, the four-copy configuration has 2338 allowed colorings. The rational quadratic expression in `code/verify_planar_centered_square_one.py` has minimum 0 on these masks.

For the corresponding kernel, write

$$H(t) = K(t) - 600J_0(t).$$

Interval evaluation with grid spacing $1/20$ on $[0, 125]$ gives $H(t) < 46$, and the standard Bessel tail estimate gives $|H(t)| < 46$ for $t \geq 125$.

For P_5 , the local restrictions leave 42 binary colorings. Substituting into the shifted integer polynomial gives minimum 0.

q	$ m_q $	A_q
1	33	-489
3	15	1035
4	14	-669
7	13	600
9	12	-700
13	11	548
16	10	-355
21	9	441
25	8	-105
31	7	339
36	6	-114
43	5	-139
49	4	16
57	3	-90

TABLE 2. Distance classes and coefficient sums for H_R .

A.2. Bounded-interval verification for the (ℓ_2, ℓ_7) kernel. The distance classes contributing to H_R are listed below. In the table, the number of unordered pairs at squared distance q , and A_q is the sum of their coefficients. The two remaining squared distances, 64 and 73, have coefficient sum zero and are omitted from H_R .

To verify the estimate on the bounded interval, let

$$\mathcal{Q} = \{1, 3, 4, 7, 9, 13, 16, 21, 25, 31, 36, 43, 49, 57\}, K(t) = -319 + \sum_{q \in \mathcal{Q}} A_q J_0(\sqrt{q}t),$$

where the values A_q are listed in Table 2. Evaluate the Bessel series using interval bounds with rational endpoints successively at rational left endpoints. If K_i^+ is the resulting rational upper bound for $K(t_i)$, choose the next rational endpoint so that $K_i^+ < 0$ and $D(t_{i+1} - t_i) \leq -K_i^+/2$, where the rational derivative bound is

$$D = \frac{35046648006}{3000000}.$$

The computer calculation reaches $t_L = 100$ and gives $K_i^+ < 0$ at every left endpoint. Hence $|K'| \leq D$ implies

$$K(t) \leq K_i^+ + D(t - t_i) \leq \frac{K_i^+}{2} < 0 \text{ for } t_i \leq t \leq t_{i+1},$$

so $K(t) < 0$ on $[0, 100]$.

For the large- t estimate, rational lower bounds for the fourth roots give

$$\sum_{q \in \mathcal{Q}} |A_q| q^{-1/4} < 3600.$$

For the other verifications involving J_0 on bounded intervals, rational interval evaluation at the grids below give the stated finite bounds.

kernel	range	grid spacing	finite bound
H_g	$[0, 25]$	1/400	$H_g(t_j) \leq 180$
H_s	$[0, 25]$	1/400	$H_s(t_j) < 46019/250$
K_5	$[0, 12]$	1/200	$K_5(t_j) < 389$

TABLE 3. Bounds from the verifications involving J_0 on bounded intervals.

The bounds in the table extend to the full intervals by the derivative estimates in the proof sections. The rational interval evaluations use the Bessel-series bound (2.5) and the analogous spherical series estimate in Lemma 10.

kernel	interval	grid spacing	bound at grid points	derivative
H_{34}	$[0, 6]$	1/5000	$H_{34}(t_j) < 69/100$	$ H'_{34} \leq 7/2$
H_{35}	$[0, 25]$	1/2000	$H_{35}(t_j) < 1469/500$	$ H'_{35} \leq 380/(3\pi) < 95/2$
H_{36}	$[0, 10]$	1/10000	$H_{36}(t_j) < 399/1000$	$ H'_{36} \leq 95/8$
H_{44}	$[0, 25]$	1/2500	$H_{44}(t_j) < 887/1000$	$ H'_{44} \leq 32/\pi < 12$

TABLE 4. Sample-point bounds and analytic derivative bounds for the four line kernels.

The LPs only choose the coefficients. The proofs use the rounded rational or integer data recorded in Appendix B and the arithmetic verifications described above.

A.3. Computational verification for the square–tetrahedron configuration.

The geometric and local counts in Section 9 follow from arithmetic in $\mathbb{Q}(\sqrt{3})$. Represent $a + b\sqrt{3}$ by (a, b) , with addition and multiplication given by

$$(a, b) + (c, d) = (a + c, b + d), (a, b)(c, d) = (ac + 3bd, ad + bc).$$

For the labelled points $p_0, \dots, p_{19}$, compute $d_{ij} = |p_i - p_j|^2$. For each $I \in \binom{\{0, \dots, 19\}}{4}$, determine whether the six values d_{ij} with $i, j \in I, i < j$, form the multiset $\{1, 1, 1, 1, 2, 2\}$. If the two pairs with value 2 are disjoint, the subset is a square. If they meet, it is a copy of $\mathcal{T}$. The resulting counts are

quantity	count
distinct points	20
four-point subsets with this distance pattern	46
squares	22
copies of $\mathcal{T}$	24

by Lemma 36.

The group $G = \langle \tau, \sigma \rangle$ has order four. Its action on the unordered pairs is obtained from

$$\mathcal{O}(i, j) = \left\{ \{ \min(g(i), g(j)), \max(g(i), g(j)) \} : g \in G \right\},$$

and a calculation gives the following orbit-size distribution:

orbit size	1	2	4	total
number of orbits	2	14	40	56
pairs accounted for	2	28	160	190

The representatives and the associated distances and coefficients are listed in Appendix B.

The local restrictions leave 47946 colorings by signs. Substituting the integer coefficients from Appendix B into the local expression gives $1000\mathcal{L}_S \geq 0$ for each of them.

For the kernel verification, use the grid $t_j = j/1000$ on $[0, 40]$. Outward-rounded interval evaluation gives the following upper bounds:

range	interval upper bound
$0 \leq t_j < 5$	$-247/250$
$5 \leq t_j < 10$	-12
$10 \leq t_j < 15$	$-99/100$
$15 \leq t_j < 20$	-18
$20 \leq t_j < 25$	-23
$25 \leq t_j < 30$	-22
$30 \leq t_j < 35$	-23
$35 \leq t_j \leq 40$	-26

In particular, every sample point satisfies (8.5). The derivative estimate in the proof extends the bound to the intervening intervals.

A.4. Linear-program sizes. Table 5 records the dimensions and scalar constraint counts of the displayed linear programs. The columns labelled local, kernel, and variable bounds count scalar inequalities. The equality column counts scalar equalities. Total constraints is their sum. A two-sided variable bound contributes two scalar inequalities.

LP	variables	local	kernel	equalities	variable bounds	total constraints
(ℓ_2, ℓ_7)	$172 = 1 + 18 + 153$	751	40001	0	0	40752
(ℓ_2, ℓ_2, ℓ_3)	$31 = 1 + 4 + 13 + 13$	304	20002	0	0	20306
(ℓ_3, ℓ_4)	$11 = 1 + 1 + 2 + 7$	12	30001	1	22	30036
(ℓ_3, ℓ_5)	$49 = 1 + 1 + 6 + 41$	73	50001	1	98	50173
(ℓ_3, ℓ_6)	$50 = 1 + 1 + 5 + 43$	78	100001	1	100	100180
(ℓ_4, ℓ_4)	$19 = 1 + 4 + 14$	26	62501	1	38	62566
(ℓ_2, P_5)	$47 = 2 + 9 + 36$	42	2140	0	94	2276
$S\text{-}\mathcal{T}$	$58 = 1 + 1 + 56$	47946	40001	0	114	88061

TABLE 5. Dimensions and scalar constraints of the displayed linear programs.

APPENDIX B. COEFFICIENT DATA

This appendix records the coefficient values used in the finite local inequalities. Together with the definitions and computational verifications in Appendix A, these data make the finite inequalities independent of the LPs.

The ordering conventions are fixed as follows. In the nine-point configuration, order point labels lexicographically and take the lexicographically least label in each vertex orbit, or the lexicographically least ordered pair in each unordered pair orbit, as its representative. The table rows follow that order. In the eighteen-point configuration, use the point order $00, 01, \dots, 08, 10, 11, \dots, 18$. The rows and columns of each coefficient array are indexed by $0, 1, \dots, 8$, and the squared-distance table is in increasing order.

For the square–tetrahedron configuration, the orbit $\mathcal{O}_j$ is the orbit represented in row j of the table below, and that row order defines the indices used in X_j , a_j , and B_q .

B.1. The four-copy square–center inequality. The dihedral symmetries of the unit square act on the twenty-point configuration in the main text. To avoid repeated denominators, write

$$10^6 \mathcal{F}_{W_4(1)}(x) = C + \sum_i A_i x_i + \sum_{i < j} B_{ij} x_i x_j.$$

The tables below give C , A_i , and B_{ij} as integers, with $C = -24994229$. The linear coefficients are constant on the following three vertex orbits.

indices	orbit size	A_i
0, 5, 10, 15	4	3167237
1, 2, 7, 8, 13, 14, 16, 19	8	5837166
3, 4, 6, 9, 11, 12, 17, 18	8	10000000

TABLE 6. Linear coefficients for the four-copy square–center inequality.

For the coefficients for pairs, the first column gives the representative of a pair orbit under these symmetries, the second gives its size, and the last gives the common value of β_{ij} on that orbit. These rows determine the complete upper-triangular matrix (β_{ij}) .

representative	orbit size	B_{ij}
(0, 1)	8	3129119
(0, 3)	8	−10000000
(0, 5)	4	10000000
(0, 6)	8	−2904617
(0, 7)	8	355813
(0, 8)	8	−2904617
(0, 9)	8	−618433
(0, 10)	2	10000000
(0, 11)	8	−487123
(0, 13)	8	9381567
(1, 2)	4	10000000
(1, 3)	8	−7095383
(1, 4)	8	10000000
(1, 6)	8	10000000
(1, 7)	8	0
(1, 8)	4	−8479162
(1, 9)	8	−487123
(1, 11)	8	−487123
(1, 12)	8	−487123
(1, 13)	4	4294145
(1, 14)	4	8325668
(1, 16)	4	10000000
(1, 17)	8	0
(1, 18)	8	2286183
(3, 4)	4	10000000
(3, 6)	4	355813
(3, 9)	8	487123
(3, 11)	4	0
(3, 12)	4	131310
(3, 18)	4	10000000

TABLE 7. Coefficients for pairs in the four-copy square–center inequality.

q	$ \sum_{ v_i-v_j ^2=q} \beta_{ij} $
$\frac{1}{2}$	$-\frac{6870881}{125000}$
1	360
$\frac{3}{2} + \sqrt{2}$	$-\frac{2904617}{125000}$
$\frac{3}{2}$	$-\frac{13131}{6250}$
$\frac{3}{2} - \sqrt{2}$	$-\frac{2904617}{125000}$
2	$-\frac{2541253}{62500}$
$\frac{5}{2} + \sqrt{2}$	$-\frac{487123}{125000}$
$\frac{5}{2} - \sqrt{2}$	$-\frac{9381567}{125000}$
2	$-\frac{487123}{125000}$
$2 - \sqrt{2}$	$-\frac{125000}{539541}$
$3 - 2\sqrt{2}$	$-\frac{31250}{89953}$
3	$\frac{6250}{858829}$
$4 - 2\sqrt{2}$	$\frac{50000}{487123}$
$2 + \sqrt{2}$	$\frac{125000}{487123}$
$3 + 2\sqrt{2}$	$\frac{250000}{487123}$

 TABLE 8. Distance-grouped coefficients for the $W_4(1)$ kernel.

With the kernel bound $K(t) - 600J_0(t) < 46$, the limiting density expression is

$$\Phi(p) = -\frac{24994229}{1000000} + \frac{46341569}{250000}p - \frac{42849267}{125000}p^2.$$

Direct evaluation gives $\max_{\frac{1}{5} \leq p \leq \frac{1}{4}} \Phi(p) < 0$.

B.2. The nine-point inequality. The group acting on the nine-point configuration has four vertex orbits and thirteen orbits of unordered pairs. Each pair orbit is determined by the representative in the table by applying the four symmetries of the configuration.

representative	orbit size	coefficient c_O	representative	orbit size	squared distance	a_U	b_U
00	2	-252	00,01	4	1	-998	2
01	4	-1004	00,02	4	4	-1000	0
02	2	998	00,11	2	3	500	150
11	1	-345	00,12	4	7	1000	300
			00,22	1	12	375	200
			01,02	4	1	-446	-894
			01,10	2	1	2	502
			01,11	4	1	444	948
			01,12	2	3	662	-12
			01,20	4	3	448	0
			01,21	2	4	-264	-12
			02,11	2	1	-52	-1000
			02,20	1	4	112	0

TABLE 10. Nine-point pair-orbit coefficients.

B.3. The eighteen-point inequality. The first table gives the sum of the coefficients for pairs at each squared distance. The next table gives the vertex coefficients, followed by the three arrays containing all coefficients for pairs.

q	1	3	4	7	9	13	16	21	25	31	36	43	49	57	64	73
multiplicity m_q	33	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1
coefficient sum A_q	-489	1035	-669	600	-700	548	-355	441	-105	339	-114	-139	16	-90	0	0

TABLE 11. The squared-distance multiplicities and aggregated coefficients for pairs.

The vertex coefficients are

p	00	01	02	03	04	05	06	07	08
c_p	0	-54	-54	-16	-2	15	-46	-1	26
p	10	11	12	13	14	15	16	17	18
c_p	18	-16	-52	1	-18	-23	-44	-53	0

For the coefficients for pairs, write

$$\begin{aligned} a_{jk}^{00} &= a_{(0,j),(0,k)}, \\ a_{jk}^{11} &= a_{(1,j),(1,k)}, \\ a_{jk}^{01} &= a_{(0,j),(1,k)}. \end{aligned}$$

The two within-column upper-triangular arrays and the cross-column array are as follows. A dash denotes an unused diagonal or lower-triangular entry.

$j \setminus k$	0	1	2	3	4	5	6	7	8
0	-	-100	0	0	0	0	0	0	0
1	-	-	-100	-2	-11	-12	14	-16	11
2	-	-	-	-100	-54	-100	-58	-22	-43
3	-	-	-	-	-100	-95	-100	-60	-55
4	-	-	-	-	-	-100	-87	-92	-58
5	-	-	-	-	-	-	-100	-63	-58
6	-	-	-	-	-	-	-	-100	-42
7	-	-	-	-	-	-	-	-	100
8	-	-	-	-	-	-	-	-	-

TABLE 12. The coefficients a_{jk}^{00} for pairs in the first column.

$j \setminus k$	0	1	2	3	4	5	6	7	8
0	-	100	-38	-52	-52	-49	-43	5	0
1	-	-	100	-57	-77	-46	-13	-12	0
2	-	-	-	100	-77	-100	-57	20	0
3	-	-	-	-	100	-95	-100	-12	0
4	-	-	-	-	-	100	-60	-10	0
5	-	-	-	-	-	-	100	1	0
6	-	-	-	-	-	-	-	100	0
7	-	-	-	-	-	-	-	-	100
8	-	-	-	-	-	-	-	-	-

TABLE 13. The coefficients a_{jk}^{11} for pairs in the second column.

$j \setminus k$	0	1	2	3	4	5	6	7	8
0	-100	0	0	0	0	0	0	0	0
1	-100	-100	40	14	31	30	57	26	0
2	77	-100	-100	99	78	79	100	59	0
3	63	85	-100	-100	93	67	68	26	0
4	63	57	100	-100	-100	84	56	28	0
5	52	61	65	70	-100	-100	85	7	0
6	67	96	100	73	100	-89	100	39	0
7	-85	93	90	58	60	88	100	100	0
8	-90	-80	63	47	60	60	75	100	100

TABLE 14. The coefficients a_{jk}^{01} for pairs between the two columns.

B.4. The pentagon inequality. The point labels are ordered as in the finite pentagon configuration in the main text. The linear coefficient vector is

$$(\alpha_0, \dots, \alpha_8) = (-23, 77, -100, -23, -23, 100, 100, 100, 77).$$

The nonzero coefficients for pairs are given by the following upper-triangular matrix. Blank entries are zero.

$i \backslash j$	0	1	2	3	4	5	6	7	8
0		-100	23	23	-100	0	0	0	23
1			-100	23	23	-100	-100	-100	-100
2				-100	23	77	-100	77	100
3					-100	0	0	-100	23
4						0	0	0	-100
5							-100	-100	-100
6								-48	-100
7									100
8									

TABLE 15. Coefficients for pairs in the finite pentagon inequality.

B.5. The four line inequalities. The sequence labels in the following tables are binary. For ϕ_{34} , ϕ_{35} , and ϕ_{36} they are the values of the red indicator r_j . For ϕ_{44} , they are the values $\epsilon_j = r_j = (s_j + 1)/2$. When the local weight is evaluated, the same sequence is converted to signed values by $s_j = 2\epsilon_j - 1$. Thus the listed table omits the forbidden sequences, and no other sequence is part of the domain of a potential.

The two short tables of potential values are

sequence	000	001	010	011	100	101	110
ϕ_{34}	0	-1	-2	-2	-3	-3	-3

TABLE 16. Potential values for the (ℓ_3, ℓ_4) inequality.

seq.	ϕ_{44}	seq.	ϕ_{44}	seq.	ϕ_{44}	seq.	ϕ_{44}
0001	0	0010	-8	0011	-10	0100	-12
0101	-14	0110	-14	0111	-12	1000	-12
1001	-14	1010	-14	1011	-12	1100	-10
1101	-8	1110	0				

TABLE 17. Potential values for the (ℓ_4, ℓ_4) inequality.

The two longer tables of potential values follow.

sequence	ϕ_{35}	sequence	ϕ_{35}
000010	-2	000011	0
000100	-8	000101	-8
000110	-6	001000	-14
001001	-14	001010	-14
001011	-14	001100	-13
001101	-12	010000	-20
010001	-20	010010	-20
010011	-20	010100	-22
010101	-20	010110	-20
011000	-24	011001	-22
011010	-20	011011	-18
100001	-26	100010	-26
100011	-26	100100	-30
100101	-28	100110	-28
101000	-28	101001	-28
101010	-26	101011	-26
101100	-29	101101	-27
110000	-30	110001	-30
110010	-28	110011	-28
110100	-26	110101	-26
110110	-24		

TABLE 18. Potential values for the (ℓ_3, ℓ_5) inequality.

sequence	ϕ_{36}	sequence	ϕ_{36}
000001	0	000010	-2
000011	-2	000100	-4
000101	-4	000110	-4
001000	-6	001001	-6
001010	-6	001011	-6
001100	-6	001101	-6
010000	-8	010001	-8
010010	-8	010011	-8
010100	-9	010101	-8
010110	-9	011000	-8
011001	-8	011010	-8
011011	-8	100000	-10
100001	-10	100010	-11
100011	-10	100100	-12
100101	-11	100110	-13
101000	-11	101001	-11
101010	-11	101011	-10
101100	-12	101101	-11
110000	-10	110001	-10
110010	-10	110011	-10
110100	-10	110101	-10
110110	-10		

TABLE 19. Potential values for the (ℓ_3, ℓ_6) inequality.

B.6. The twenty-point square–tetrahedron inequality. For each row, the representative is a pair of point labels from the square–tetrahedron section. The third column is the size of its orbit under the group G , the fourth is the common squared distance, and the last is the coefficient a_j in the square–tetrahedron expression.

q	B_q	q	B_q
$2 - \sqrt{3}$	-68300	1	-300708
$3 - \sqrt{3}$	56588	2	-142872
$4 - \sqrt{3}$	-14632	3	18334
$5 - \sqrt{3}$	10996	$2 + \sqrt{3}$	628
4	-5700	$3 + \sqrt{3}$	-404
$4 + \sqrt{3}$	204	6	386
$5 + \sqrt{3}$	380	7	136

TABLE 20. Aggregated squared-distance coefficients for the square–tetrahedron kernel.

TABLE 21. The integer pair-orbit data for the square–tetrahedron result.

j	representative	$ \mathcal{O}_j $	q_j	a_j	j	representative	$ \mathcal{O}_j $	q_j	a_j
0	$\{8, 10\}$	4	1	0	28	$\{1, 8\}$	4	1	-2603
1	$\{0, 8\}$	4	1	-17344	29	$\{3, 5\}$	4	2	-2761
2	$\{0, 10\}$	2	2	-152	30	$\{3, 15\}$	4	$4 - \sqrt{3}$	-3658
3	$\{0, 1\}$	4	1	-9839	31	$\{1, 10\}$	4	$2 - \sqrt{3}$	-337
4	$\{4, 8\}$	4	2	-21087	32	$\{3, 16\}$	2	6	193
5	$\{0, 2\}$	4	1	-10680	33	$\{7, 15\}$	4	$5 - \sqrt{3}$	2749
6	$\{4, 10\}$	2	3	587	34	$\{7, 16\}$	2	7	68
7	$\{0, 3\}$	4	2	-2319	35	$\{2, 10\}$	4	$2 + \sqrt{3}$	-294
8	$\{3, 4\}$	4	3	2040	36	$\{3, 9\}$	4	$2 - \sqrt{3}$	-6010
9	$\{1, 4\}$	4	2	-553	37	$\{3, 7\}$	2	1	-2480
10	$\{1, 3\}$	4	1	3317	38	$\{9, 11\}$	2	3	460
11	$\{2, 3\}$	4	1	-997	39	$\{5, 8\}$	4	2	-5565
12	$\{7, 9\}$	4	$3 - \sqrt{3}$	5100	40	$\{1, 15\}$	2	$2 - \sqrt{3}$	-7650
13	$\{0, 4\}$	1	1	-39099	41	$\{6, 8\}$	4	$3 + \sqrt{3}$	-21
14	$\{5, 14\}$	4	4	-1425	42	$\{7, 8\}$	4	$3 + \sqrt{3}$	-145
15	$\{2, 4\}$	4	2	-1277	43	$\{10, 11\}$	4	2	537
16	$\{2, 14\}$	2	$2 + \sqrt{3}$	-12	44	$\{2, 8\}$	4	$2 + \sqrt{3}$	-93
17	$\{1, 14\}$	4	3	1304	45	$\{3, 10\}$	4	2	-59
18	$\{3, 14\}$	4	$4 + \sqrt{3}$	51	46	$\{3, 6\}$	4	2	1433
19	$\{6, 14\}$	2	$3 + \sqrt{3}$	-22	47	$\{6, 9\}$	4	2	-2065
20	$\{1, 9\}$	4	$2 - \sqrt{3}$	-6903	48	$\{3, 8\}$	4	$2 + \sqrt{3}$	550
21	$\{7, 14\}$	4	$5 + \sqrt{3}$	95	49	$\{5, 15\}$	2	$3 - \sqrt{3}$	5870
22	$\{1, 2\}$	4	2	-1171	50	$\{6, 10\}$	4	$3 + \sqrt{3}$	76
23	$\{5, 9\}$	4	$3 - \sqrt{3}$	5627	51	$\{8, 9\}$	2	2	-1510
24	$\{1, 5\}$	2	1	-6775	52	$\{7, 10\}$	4	3	-428
25	$\{5, 10\}$	4	$3 - \sqrt{3}$	485	53	$\{10, 13\}$	1	1	-267
26	$\{2, 5\}$	4	3	1144	54	$\{2, 6\}$	2	1	-54
27	$\{2, 9\}$	4	1	-9068	55	$\{8, 11\}$	2	1	-26934

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REFERENCES

- [1] G. Ambrus and M. Matolcsi, *Density estimates of 1-avoiding sets via higher order correlations*, Discrete Comput. Geom. **67** (2022), no. 4, 1245–1256.
- [2] G. Ambrus, A. Csizsárik, M. Matolcsi, D. Varga, and P. Zsámboki, *The density of planar sets avoiding unit distances*, Math. Program. **207** (2024), no. 1–2, 303–327.
- [3] A. Arman and S. Tsaturian, *A result in asymmetric Euclidean Ramsey theory*, Discrete Math. **341** (2018), no. 5, 1502–1508.
- [4] C. Bachoc, A. Passuello, and A. Thiery, *The density of sets avoiding distance 1 in euclidean space*, Discrete Comput. Geom. **53** (2015), no. 4, 783–808.
- [5] K. Cantwell, *Finite Euclidean Ramsey theory*, J. Combin. Theory Ser. A **73** (1996), no. 2, 273–285.
- [6] D. Conlon and J. Fox, *Lines in Euclidean Ramsey theory*, Discrete Comput. Geom. **61** (2019), no. 1, 218–225.

- [7] D. Conlon and J. Führer, *Non-spherical sets versus lines in Euclidean Ramsey theory*, *Canad. Math. Bull.* **69** (2026), no. 1, 179–183.
- [8] D. Conlon and Y.-H. Wu, *More on lines in Euclidean Ramsey theory*, *C. R. Math. Acad. Sci. Paris* **361** (2023), 897–901.
- [9] G. Csizmadia and G. Tóth, *Note on a Ramsey-type problem in geometry*, *J. Combin. Theory Ser. A* **65** (1994), no. 2, 302–306.
- [10] G. Currier, P. Mody, Z. Xie, and J. Zhang, *Improved bounds for lines and 1-separated sets in Euclidean Ramsey theory*, arXiv:2606.17194 [math.CO], 2026.
- [11] G. Currier, K. Moore, and C. H. Yip, *Any two-coloring of the plane contains monochromatic 3-term arithmetic progressions*, *Combinatorica* **44** (2024), no. 6, 1367–1380.
- [12] G. Currier, K. Moore, and C. H. Yip, *Avoiding short progressions in Euclidean Ramsey theory*, *J. Combin. Theory Ser. A* **217** (2026), 106080.
- [13] E. DeCorte, F. M. de Oliveira Filho, and F. Vallentin, *Complete positivity and distance-avoiding sets*, *Math. Program.* **191** (2022), no. 2, 487–558.
- [14] F. M. de Oliveira Filho and F. Vallentin, *Fourier analysis, linear programming, and densities of distance avoiding sets in $\mathbb{R}^n$* , *J. Eur. Math. Soc.* **12** (2010), no. 6, 1417–1428.
- [15] P. Diaconis and D. Freedman, *A Dozen de Finetti-Style Results in Search of a Theory*, *Ann. Inst. Henri Poincaré Probab. Stat.* **23** (1987), no. S2, 397–423.
- [16] P. Erdős, R. L. Graham, P. Montgomery, B. L. Rothschild, J. Spencer, and E. G. Straus, *Euclidean Ramsey theorems. I*, *J. Combin. Theory Ser. A* **14** (1973), no. 3, 341–363.
- [17] P. Erdős, R. L. Graham, P. Montgomery, B. L. Rothschild, J. H. Spencer, and E. G. Straus, *Euclidean Ramsey theorems II*, in *Infinite and Finite Sets*, *Colloquia Mathematica Societatis János Bolyai*, vol. 10, North-Holland, Amsterdam, 1975, pp. 529–557.
- [18] P. Erdős, R. L. Graham, P. Montgomery, B. L. Rothschild, J. H. Spencer, and E. G. Straus, *Euclidean Ramsey theorems III*, in *Infinite and Finite Sets*, *Colloquia Mathematica Societatis János Bolyai*, vol. 10, North-Holland, Amsterdam, 1975, pp. 559–583.
- [19] J. Führer and G. Tóth, *Progressions in Euclidean Ramsey theory*, *European J. Combin.* **125** (2025), 104105.
- [20] W. Gasarch, A. Gezalyan, and R. Parker, *Open Problems Column Monochromatic Unit Squares: Exposition and Open Problems*, *SIGACT News* **56** (2025), no. 3, 38–55, <https://doi.org/10.1145/3767145.3767149>.
- [21] L. Iván, *Monochromatic point sets in the plane and in space*, Master’s thesis, University of Szeged, 1979.
- [22] R. Juhász, *Ramsey type theorems in the plane*, *J. Combin. Theory Ser. A* **27** (1979), no. 2, 152–160.
- [23] T. Keleti, M. Matolcsi, F. M. de Oliveira Filho, and I. Z. Ruzsa, *Better bounds for planar sets avoiding unit distances*, *Discrete Comput. Geom.* **55** (2016), no. 3, 642–661.
- [24] L. Moser and W. Moser, *Solution to problem 10*, *Canad. Math. Bull.* **4** (1961), 187–189.
- [25] F. W. J. Olver, A. B. Olde Daalhuis, D. W. Lozier, B. I. Schneider, R. F. Boisvert, C. W. Clark, B. R. Miller, B. V. Saunders, H. S. Cohl, and M. A. McClain, *NIST Digital Library of Mathematical Functions*, <https://dlmf.nist.gov/>, Release 1.2.7 of 2026-06-15.
- [26] B. Simon, *Real Analysis: A Comprehensive Course in Analysis, Part 1*, American Mathematical Society, Providence, RI, 2015.
- [27] L. A. Székely, *Measurable chromatic number of geometric graphs and sets without some distances in euclidean space*, *Combinatorica* **4** (1984), no. 2–3, 213–218.
- [28] A. D. Szlam, *Monochromatic translates of configurations in the plane*, *J. Combin. Theory Ser. A* **93** (2001), no. 1, 173–176.
- [29] S. Tsaturian, *A Euclidean Ramsey result in the plane*, *Electron. J. Comb.* **24** (2017), no. 4, Paper No. P4.35, 9 pp.

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